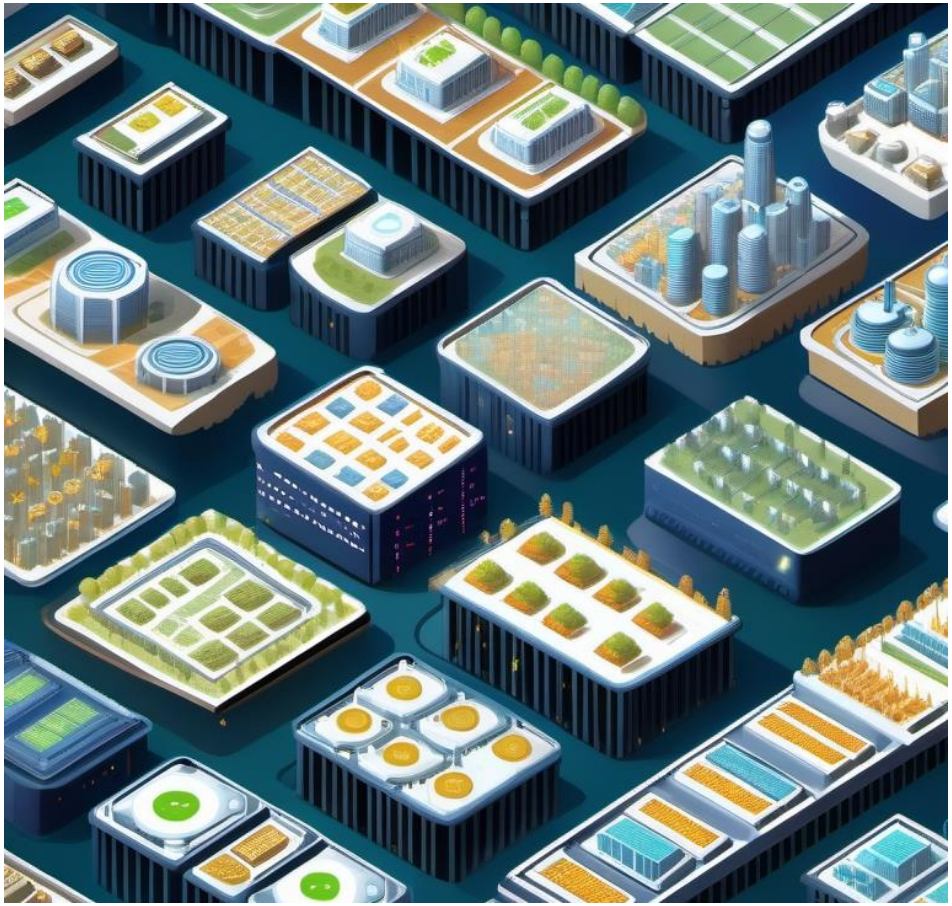


GESTIÓN DE SISTEMSA DE DATOS MASIVOS

INDEX



01. Introduction

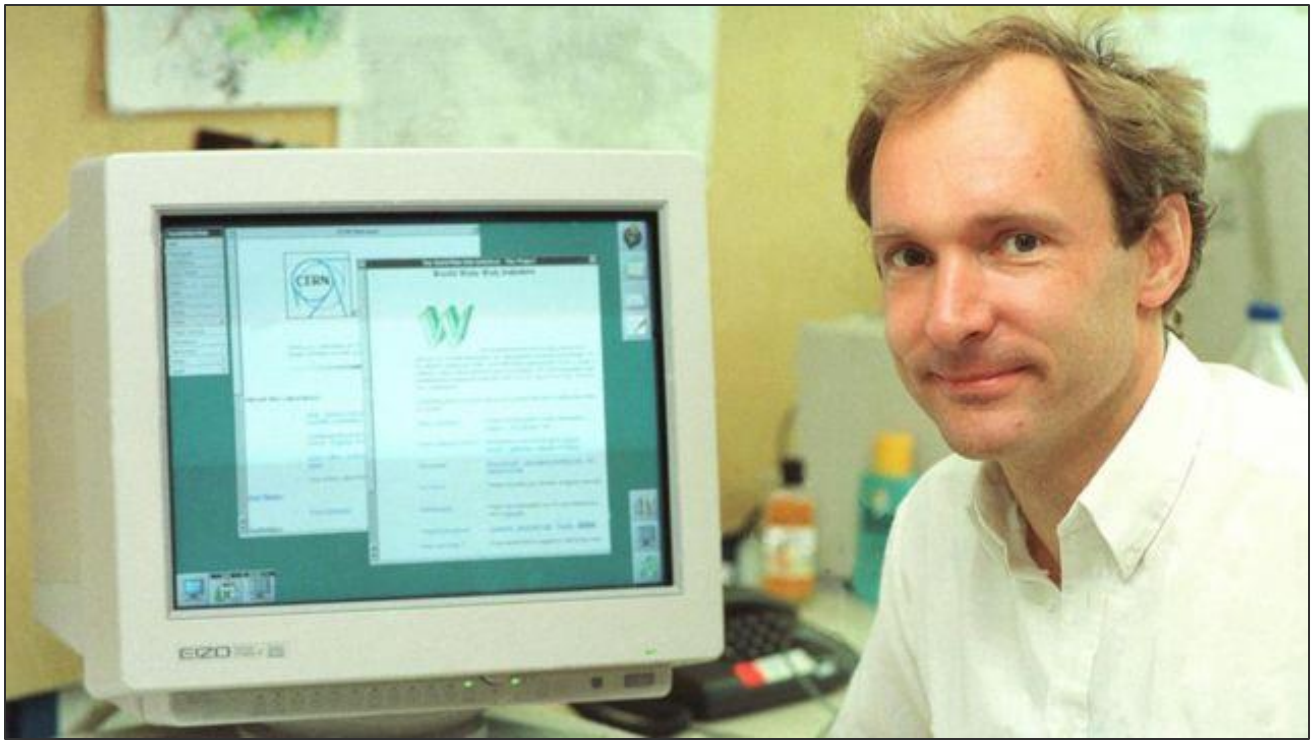
02. RDF

03. Schemas and Ontologies

04. Sample cases

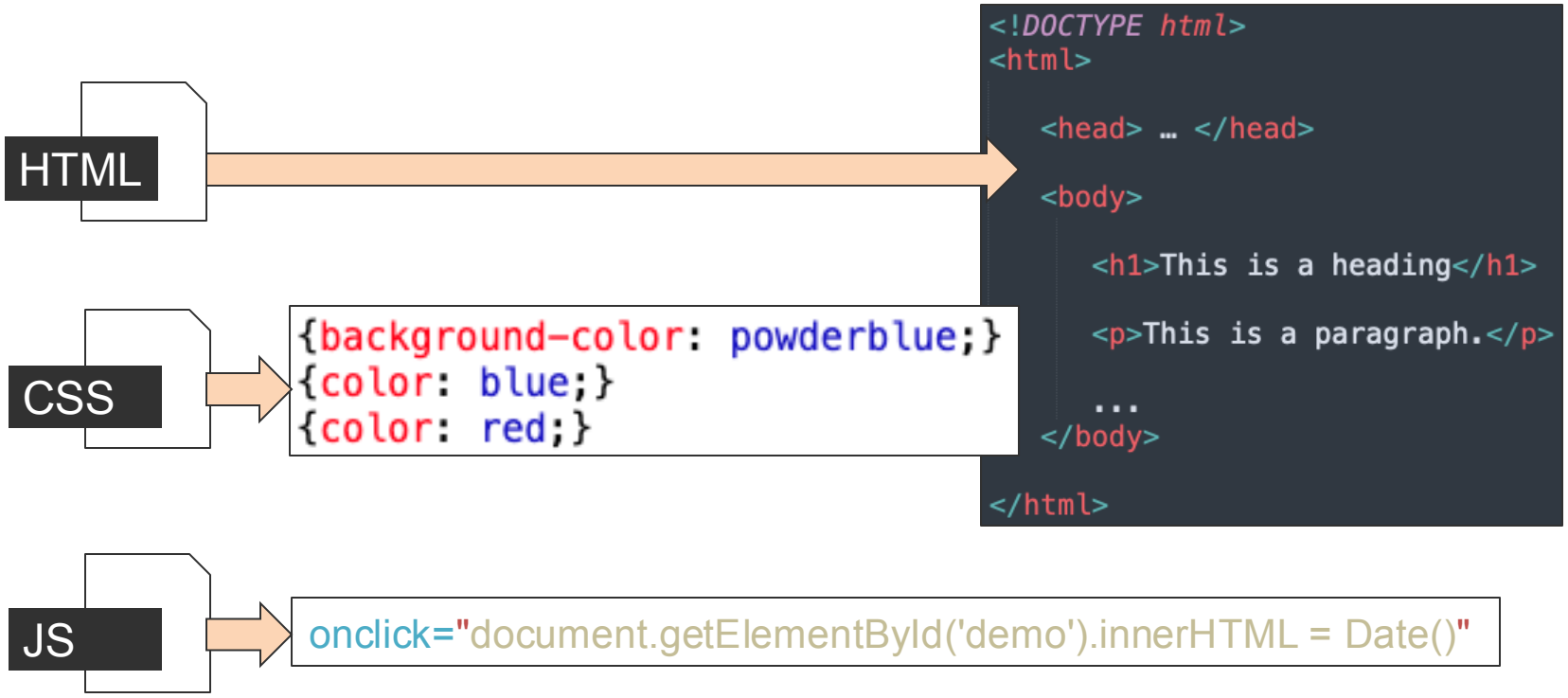
A bit of history

A long time ago in a laboratory far, far away...

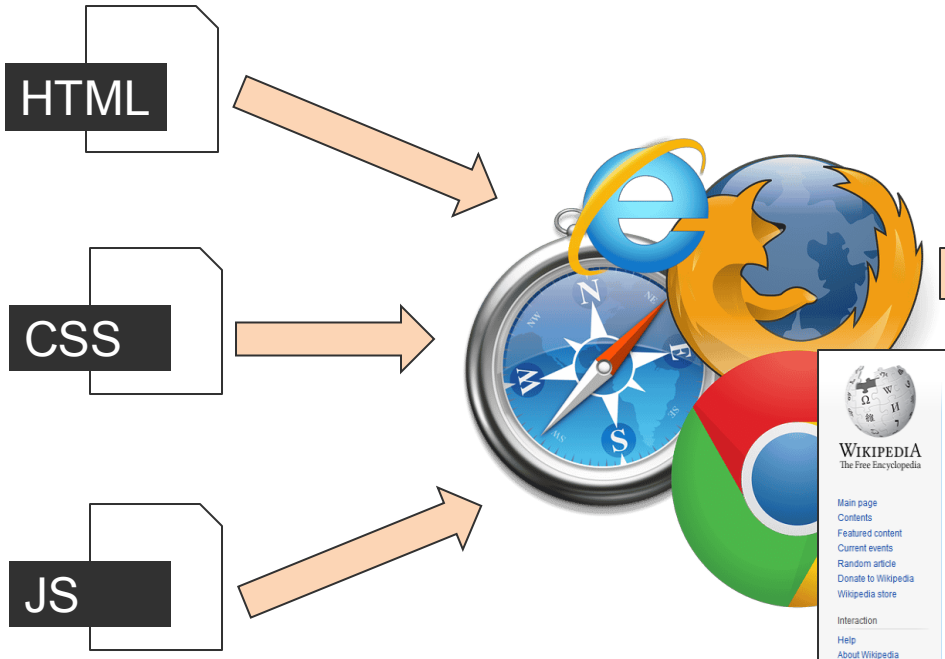


Tim Berners-Lee demonstrating the World Wide Web as a scientific tool in 1989.

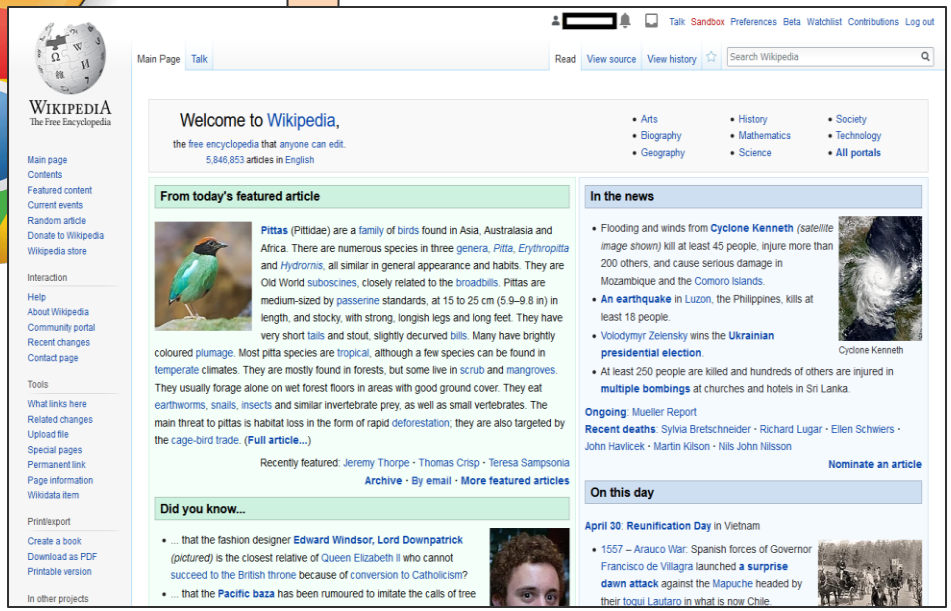
The Web at its origins...



The Web at its origins...

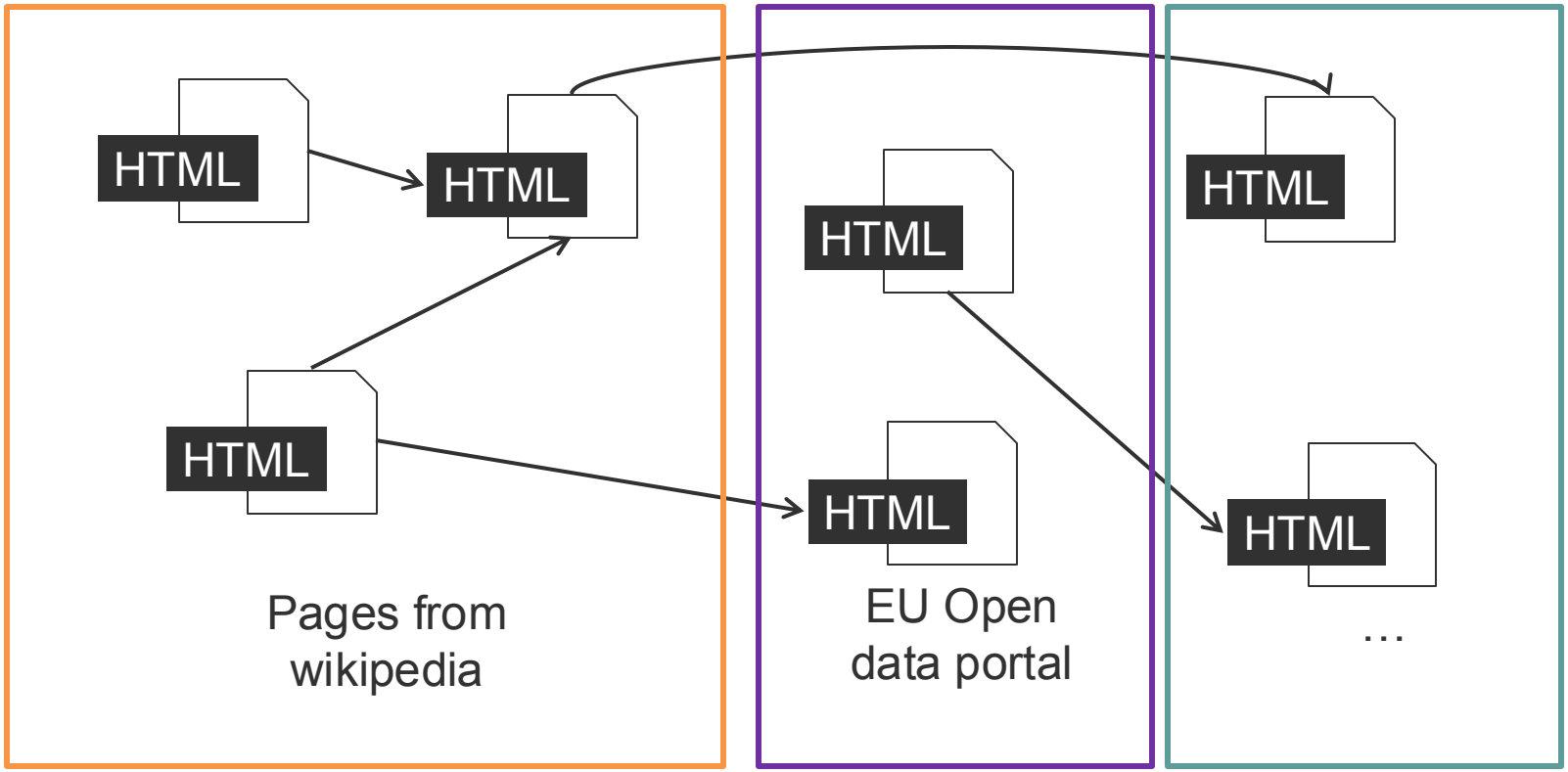


https://en.wikipedia.org/



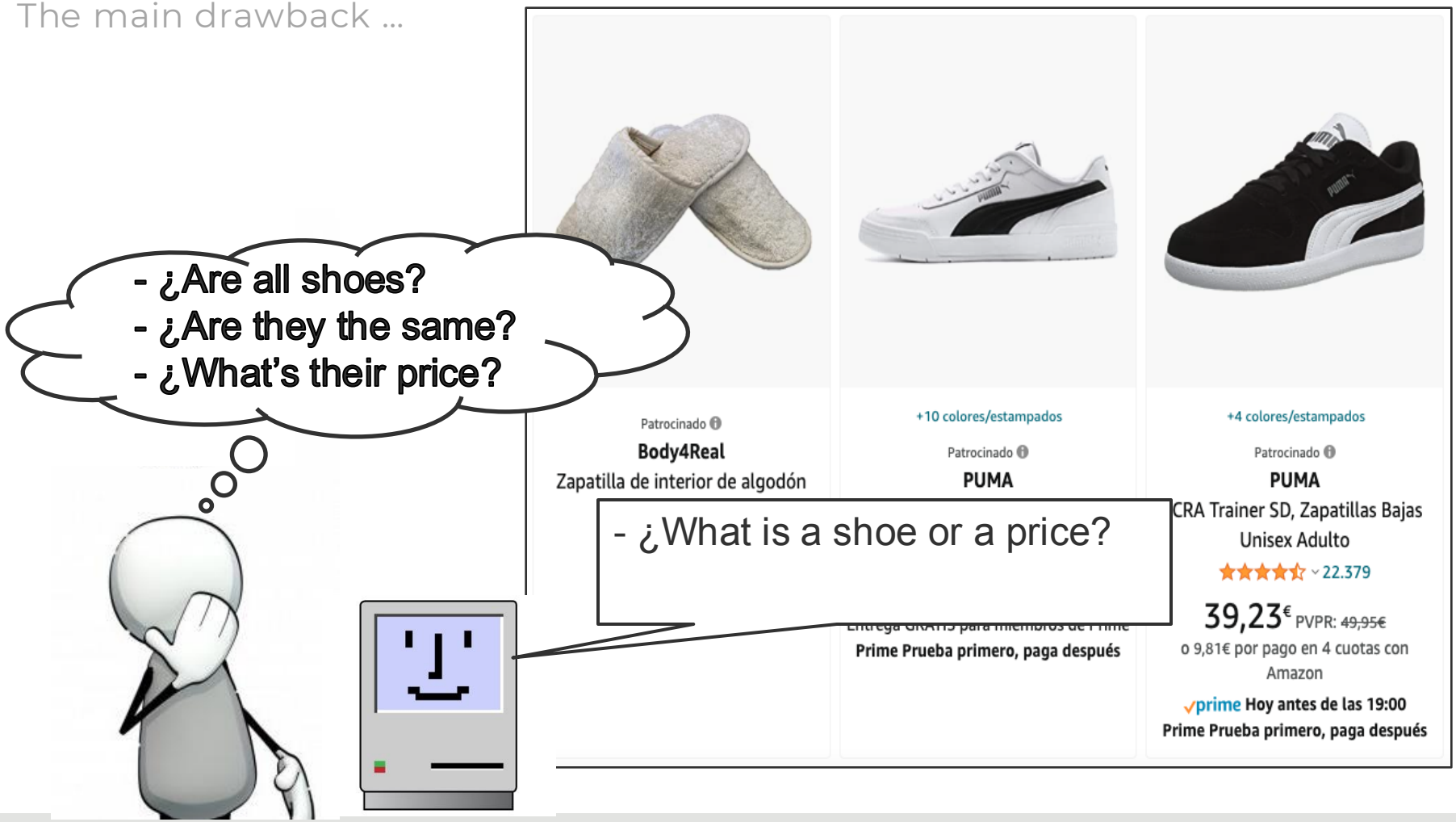
The Web at its origins...

Linked ~~data~~ documents



The Web at its origins...

The main drawback ...



The Web evolution

The main drawback ...



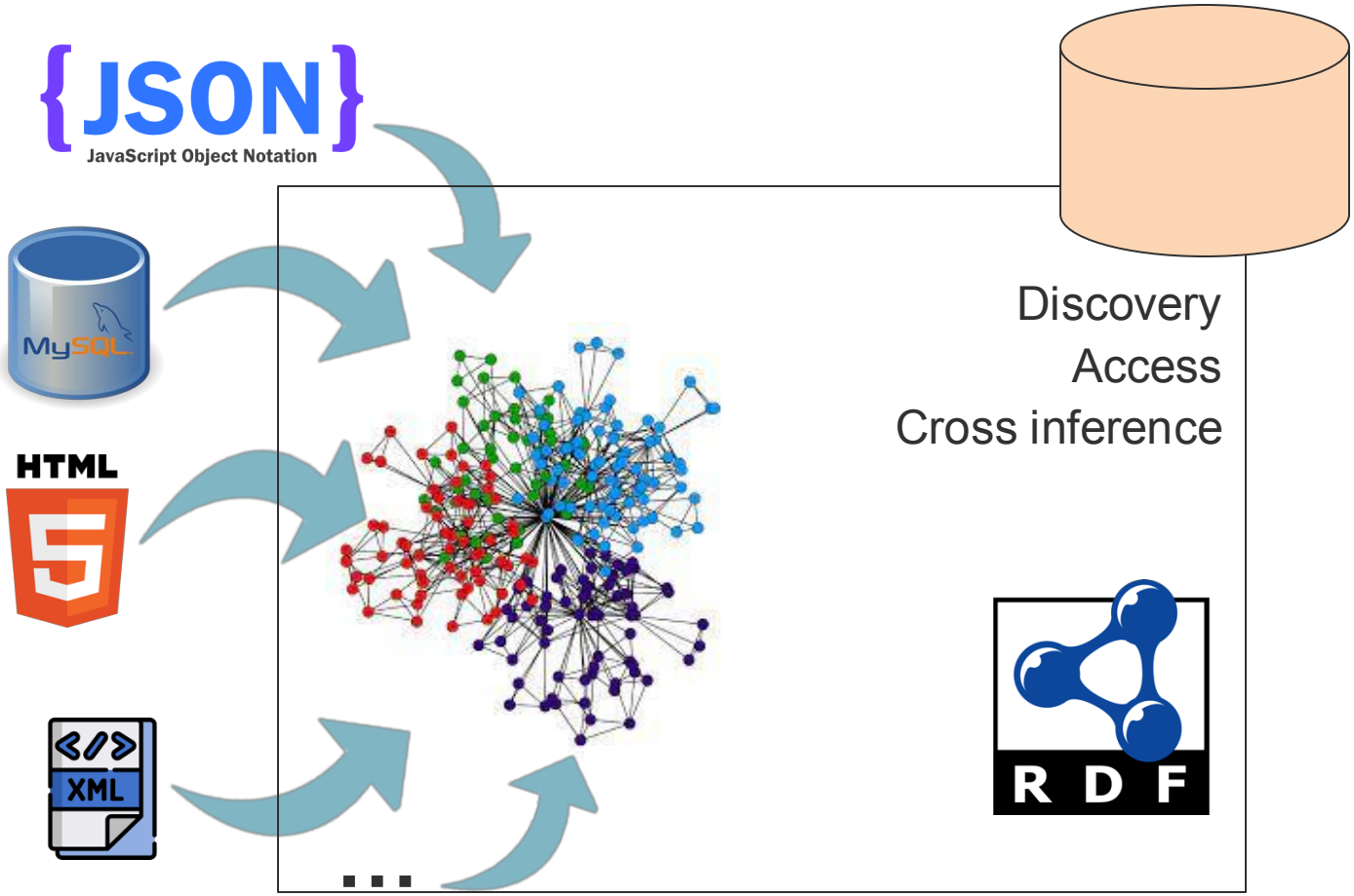
The Web evolution

A first approach to the problems...

- We will define a set of entities and their properties by assigning them an identifier so that their definition is unambiguous (schema).
- We will define the meaning of the entities and their properties so that they can be processed by both humans and machines (ontology).
- We are going to express the data in a structured form that allows us to use unique identifiers on the Web and that is consistent with the defined entities and properties (semantic data).
- We will relate the data entities that talk about entities that have something to do with each other (linked data).
- We will put the data and the definition of the entities, and their properties, on the Web so that they can be accessed.

The Web evolution

A first approach to the problems...



The Web evolution

A first approach to the problems → Birth of the Semantic web



THE SEMANTIC WEB
TIM BERNERS-LEE, JAMES HENDLER and ORA LASSILA
Scientific American
Vol. 284, No. 5 (MAY 2001), pp. 34-43 (10 pages)
Published By: Scientific American, a division of Nature America, Inc.

https://www.w3.org/standards/semanticweb/

Agents

Expressing Meaning

Evolution of Knowledge

Ontologies

Knowledge Representation

5-Star Linked Data

"Quality" principles



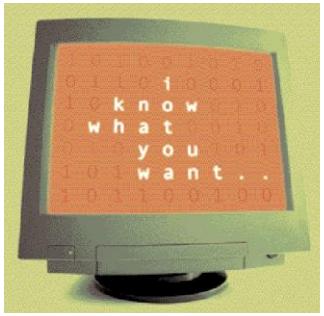
https://5stardata.info/es/

What is Semantic Web ?

"The Semantic Web is an extension of the current Web in which information is given well-defined meaning, better enabling computers and people to work in cooperation. It is based on the idea of having data on the Web defined and linked such that it can be used for more effective discovery, automation, integration, and reuse across various applications."

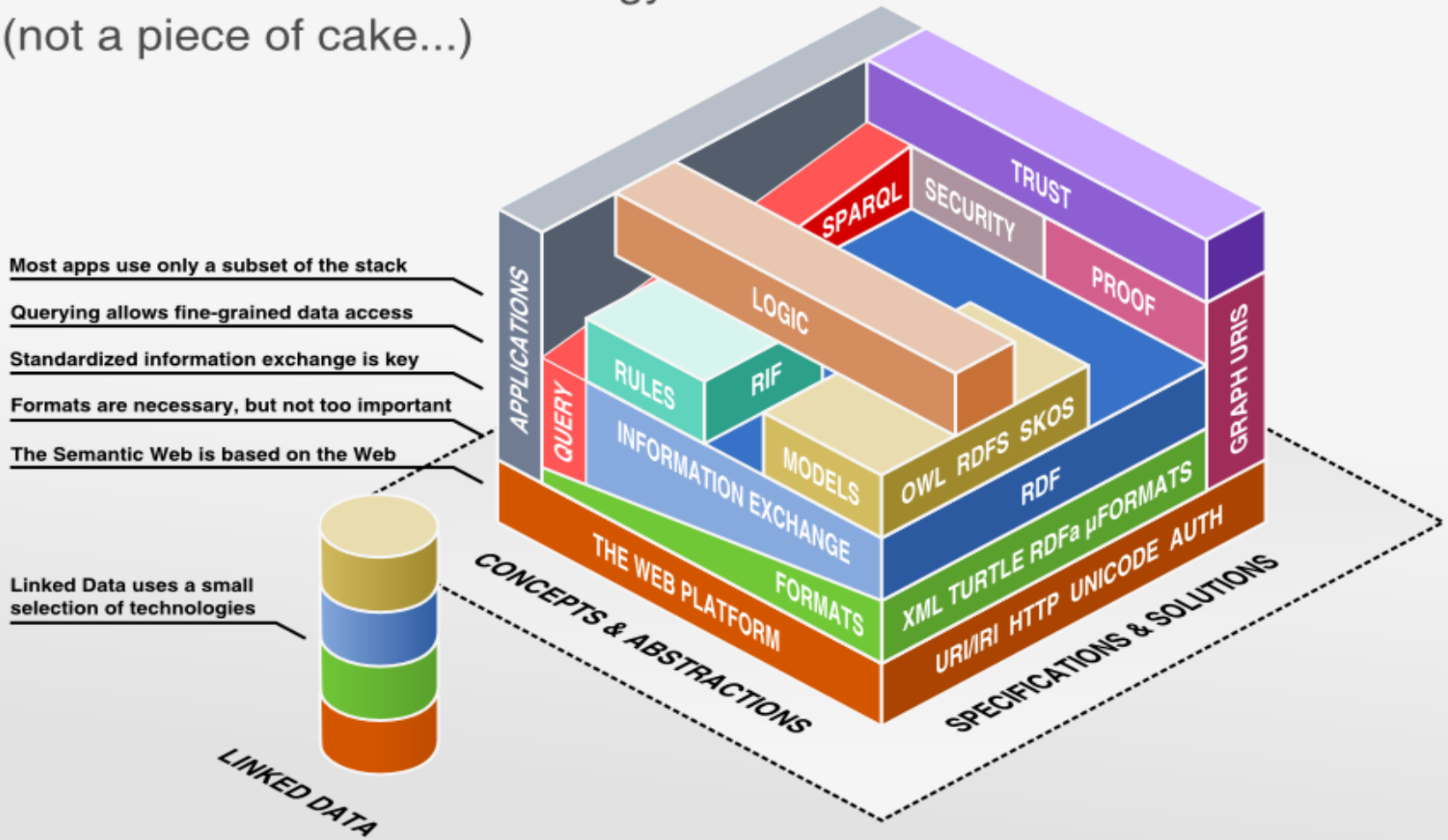
Ontologies

Annotation



Hendler, J., Berners-Lee, T., and Miller, E.
Integrating Applications on the Semantic Web, 2002,
http://www.w3.org/2002/07/swint.html

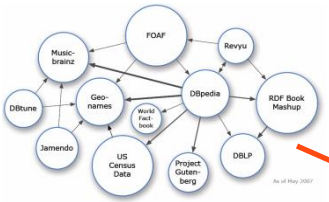
The Semantic Web Technology Stack (not a piece of cake...)



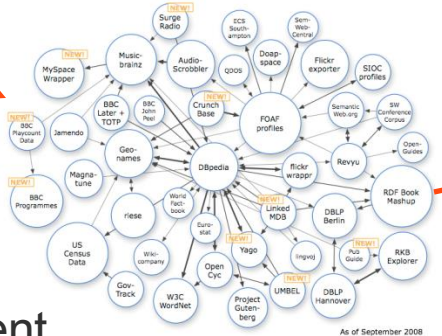
The Semantic World Wide Web



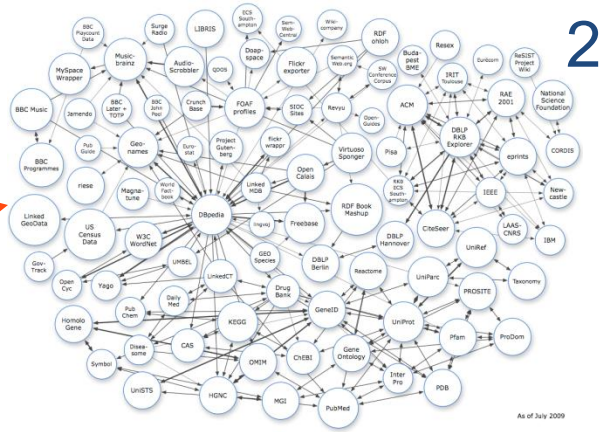
Link (Open) Data



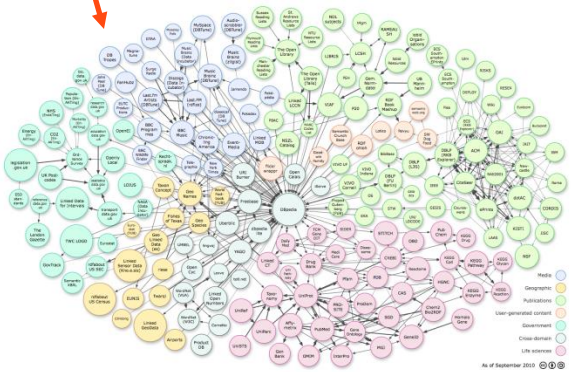
2007



2008



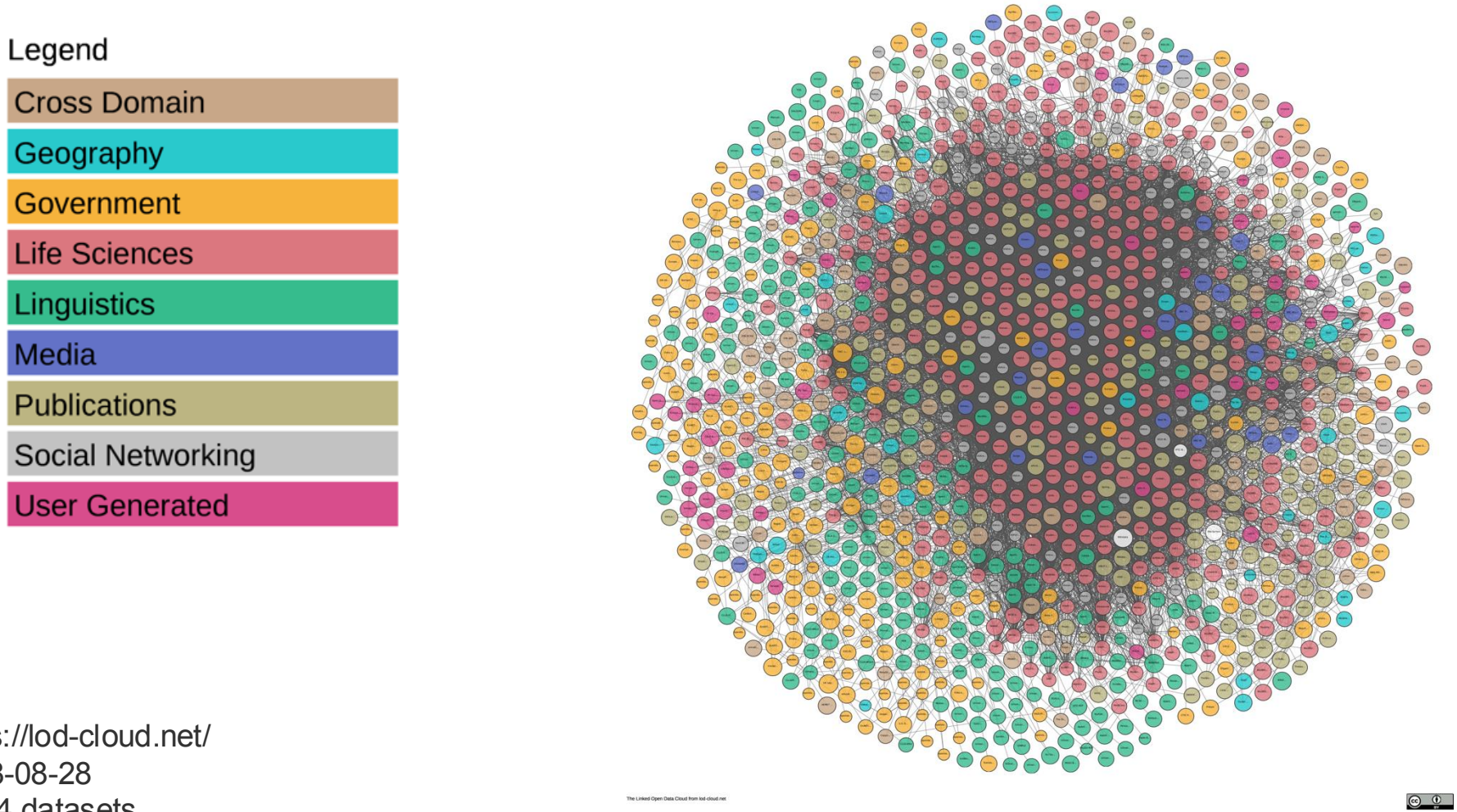
2009



2010

- UK Government
- US Government
- BBC
- Open Calais
- Freebase
- NY Times
- CNET
- DBpedia
- ....

Link (Open) Data (2018)



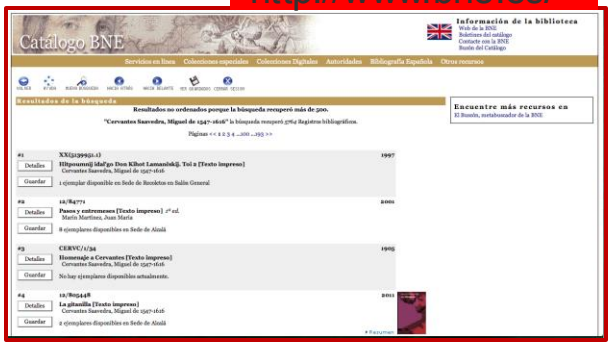
https://lod-cloud.net/
2018-08-28
1,224 datasets

Paradigm change

Complex queries over heterogeneous data sources

A person wants to know about Cervantes and wants to know more about his work, life, as well as related places to visit.

http://www.bne.es/

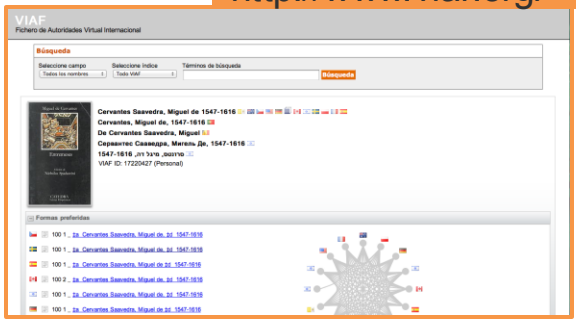




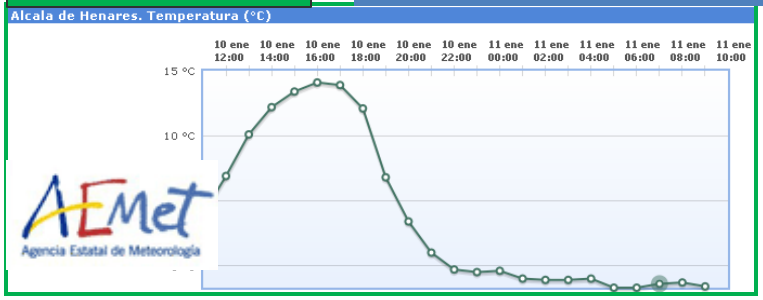
http://elviajero.elpais.com/



http://www.viaf.org/



http://www.aemet

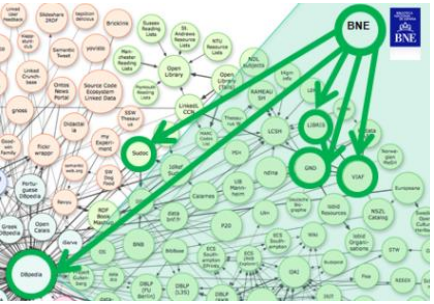


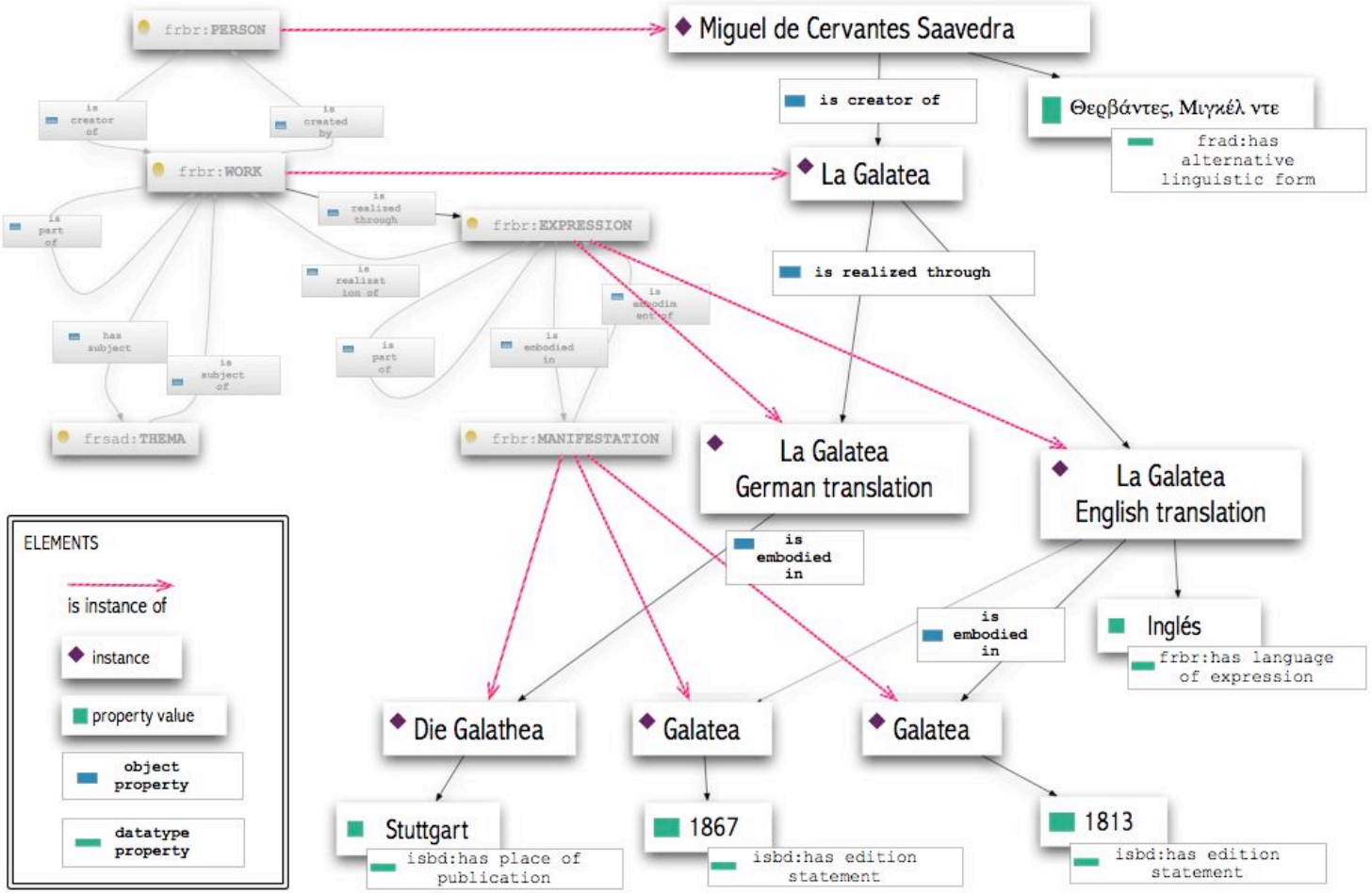




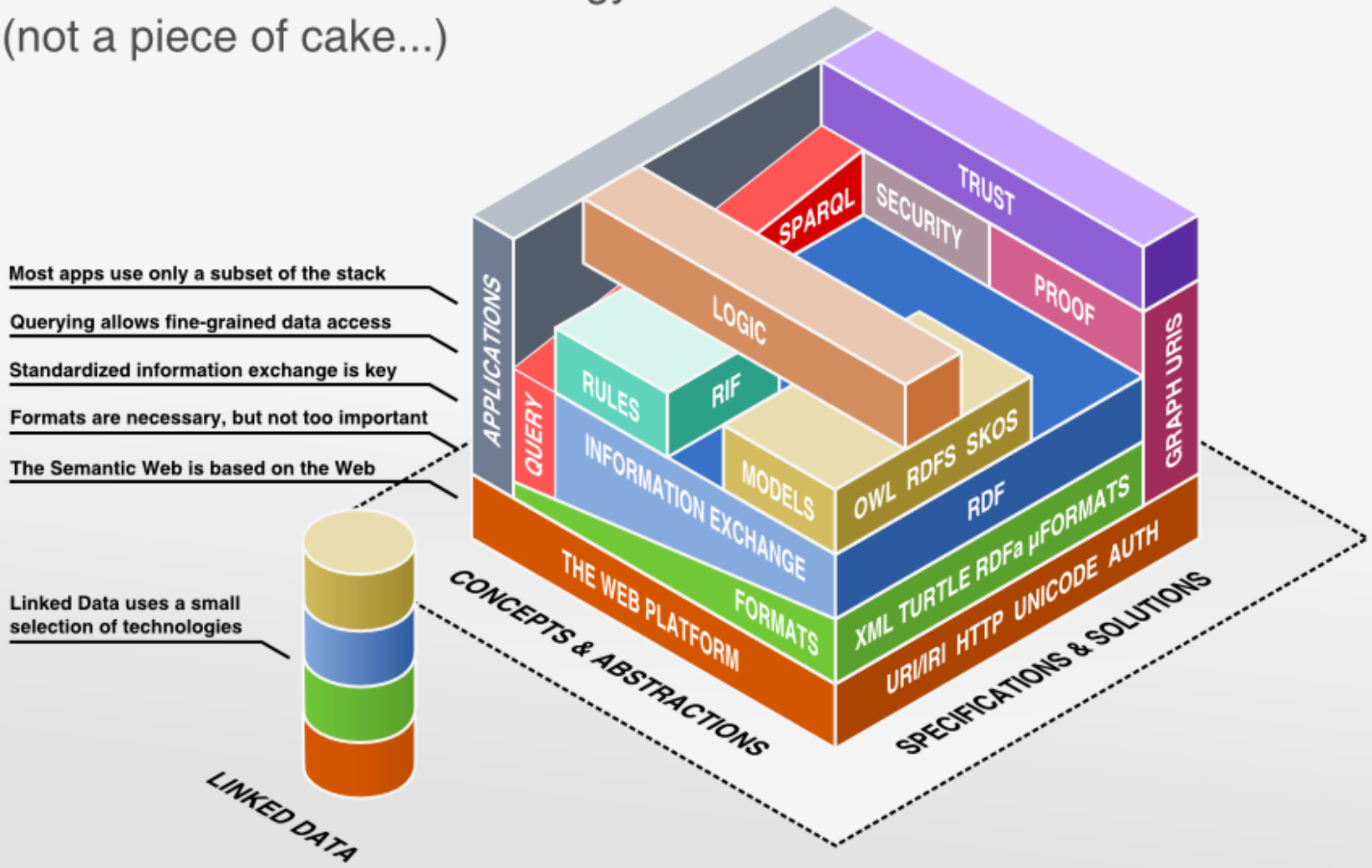


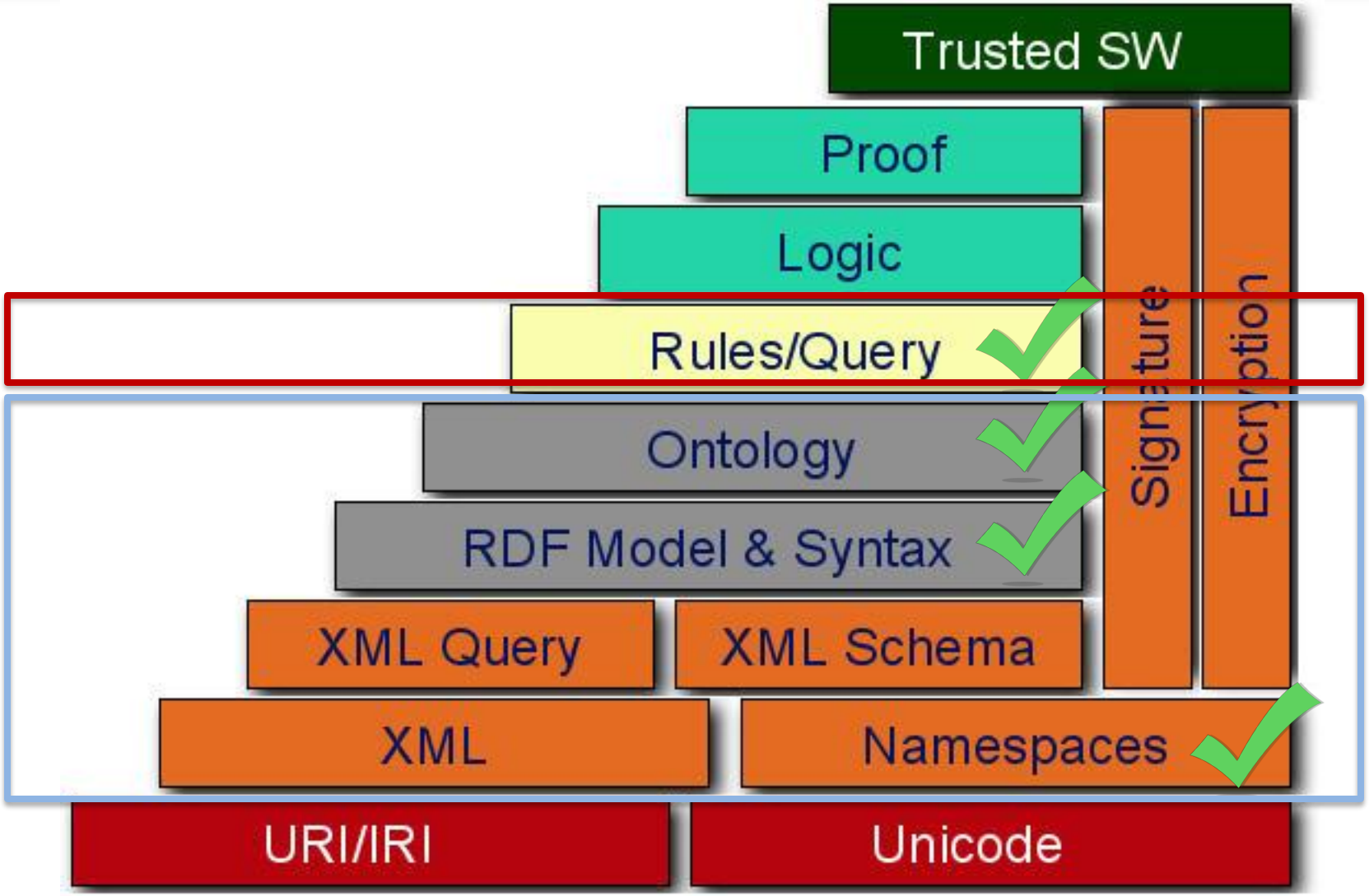
Data + Model


BNE
BNE


frbr:PERSON
Miguel de Cervantes Saavedra
is creator of
Θερβάντες, Μιγκέλ ντε
frad:has alternative linguistic form
frbr:WORK
La Galatea
frbr:EXPRESSION
is realized through
frsad:THEMA
frbr:MANIFESTATION
La Galatea German translation
La Galatea English translation
is embodied in
Inglés
frbr:has language of expression
Die Galathea
Galatea
Galatea
Stuttgart
isbd:has place of publication
1867
isbd:has edition statement
1813
isbd:has edition statement
ELEMENTS
is instance of
instance
property value
object property
datatype property

The Semantic Web Technology Stack (not a piece of cake...)




Trusted SW
Proof
Logic
Rules/Query
Ontology
RDF Model & Syntax
XML Query
XML Schema
XML
Namespaces
URI/IRI
Unicode
Signature
Encryption

Who relies on Linked Data

- RTVE a la carta
- Public administrations
  - Portales de datos abiertos (Zaragoza, Madrid, ...)
  - Gobierno (BOE)
  - Centro Nacional de Información Geográfica (CNIG)
- Cultural sites
  - Museums (Muse del prado)
  - Libraries (BNE)
  - ...
- EUROP
  - Cultural heritage (https://www.europeana.eu/es)
  - Planet views: geo, atmospheric, ... (https://www.copernicus.eu/es)
- Bussiness
  - Google for indexing web pages relies on schema.org

DATA SPACES



EUROPEAN COMMISSION

Brussels, 19.2.2020

COM(2020) 66 final

COMMUNICATION FROM THE COMMISSION TO THE EUROPEAN PARLIAMENT, THE COUNCIL, THE EUROPEAN ECONOMIC AND SOCIAL COMMITTEE AND THE COMMITTEE OF THE REGIONS

A European strategy for data

https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A52020DC0066

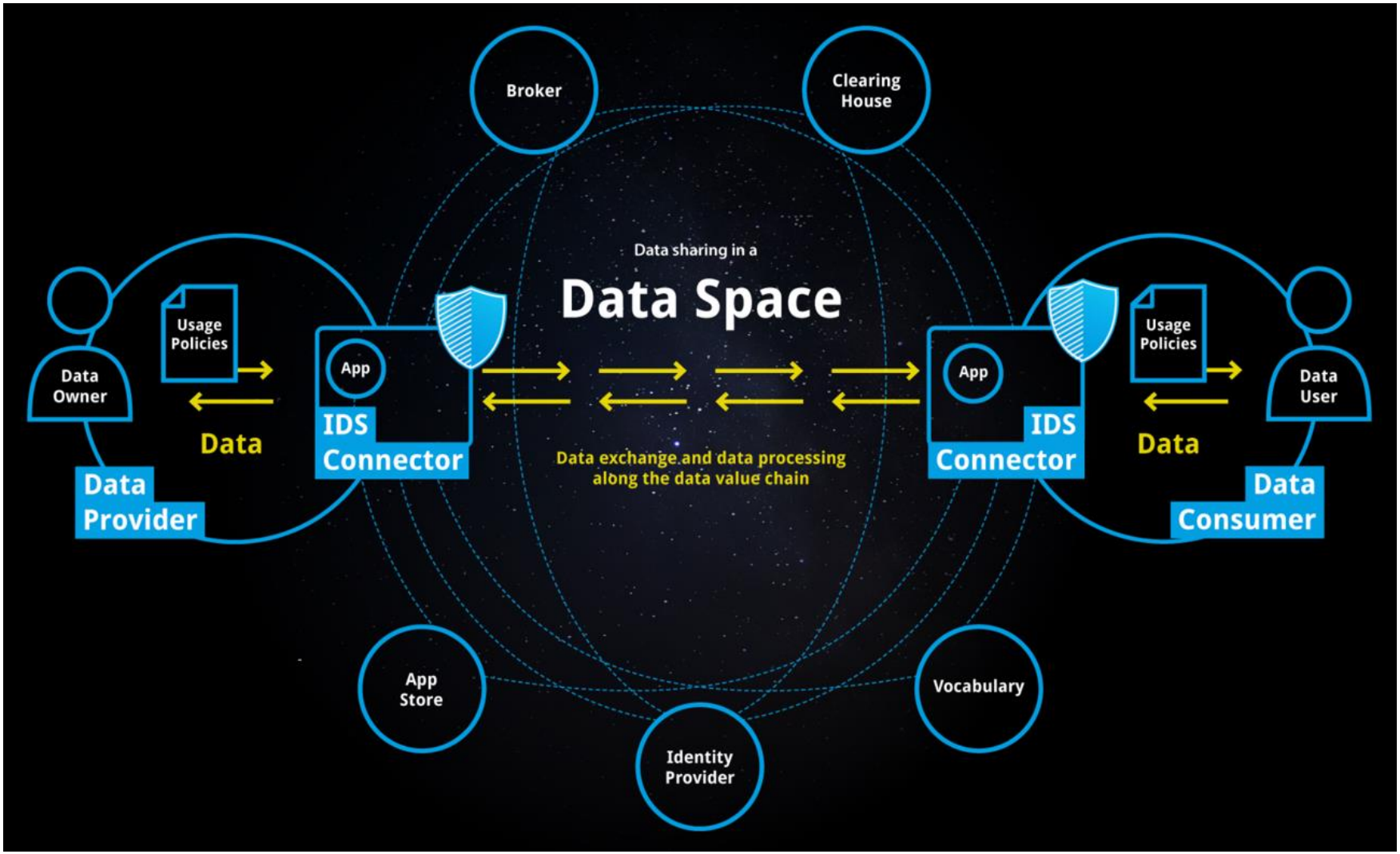


Home > Library > Staff working document on data spaces

POLICY AND LEGISLATION | Publication 14 February 2022

Staff working document on data spaces

https://digital-strategy.ec.europa.eu/en/library/staff-working-document-data-spaces


Broker
Clearing House
Data sharing in a
Data Space
Usage Policies
Data Owner
App
IDS Connector
Data
Data Provider
Data exchange and data processing along the data value chain
App
IDS Connector
Usage Policies
Data User
Data
Data Consumer
App Store
Identity Provider
Vocabulary

DATA SPACES

INTERNATIONAL DATA SPACES ASSOCIATION

https://internationaldataspaces.org/

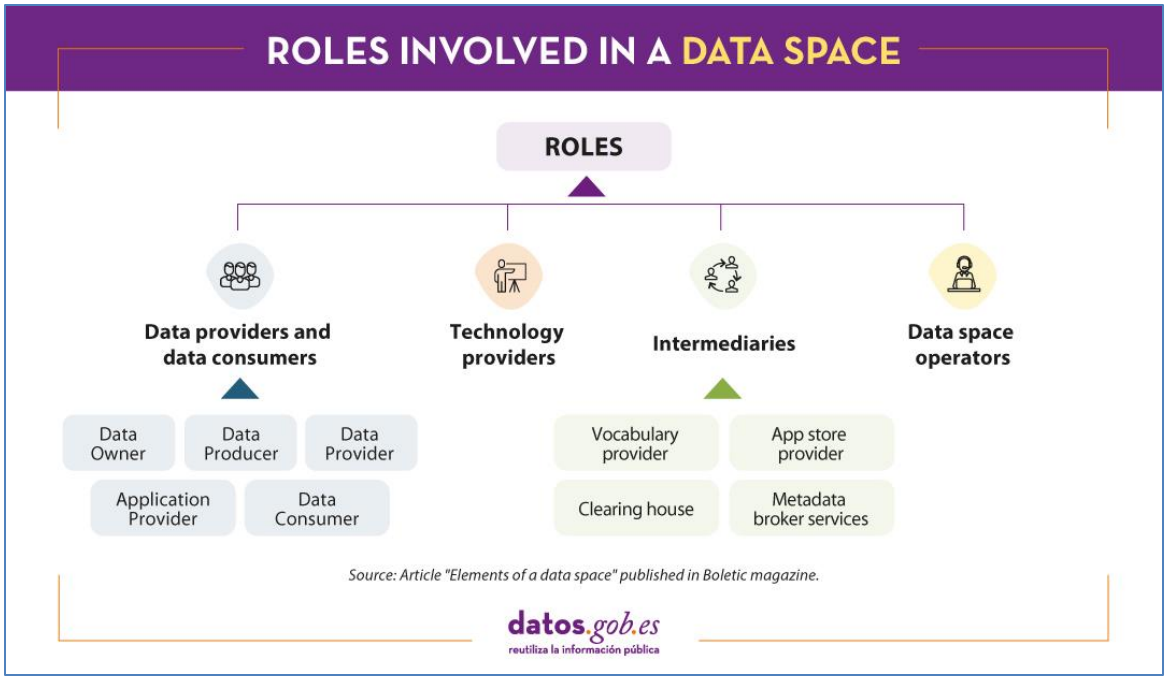
FIWARE FOUNDATION

https://www.fiware.org/foundation/

GAIA-X

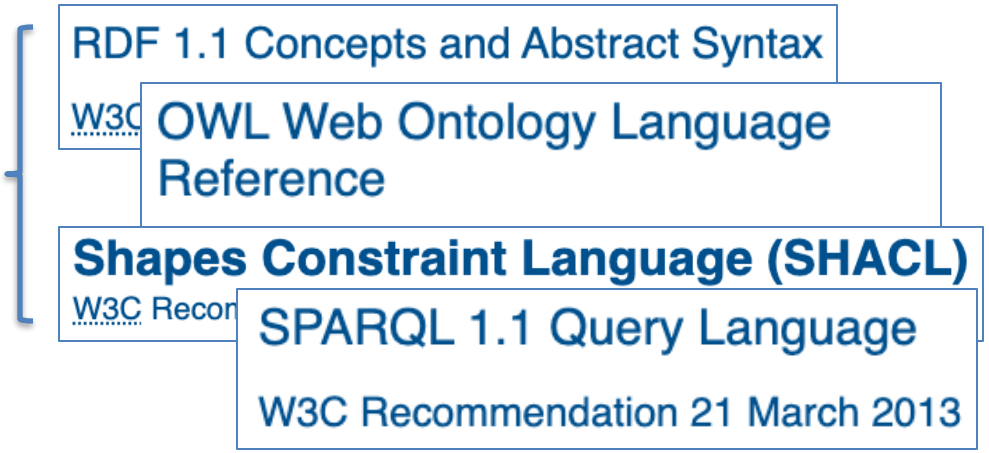
https://www.gaiax.es/

LINKED DATA & DATA SPACES



LINKED DATA & DATA SPACES

- Semantic Interoperability
  - Data formats
  - Knowledge definition
  - Data validation
- Data Catalogues
  - Data discovery
  - Meta description
- Policy-based privacy
- Identity providers
- Linked data services



LINKED DATA & DATA SPACES

- Semantic Interoperability
  - Data formats
  - Knowledge definition
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- Data Catalogues
  - Data discovery
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LINKED DATA & DATA SPACES

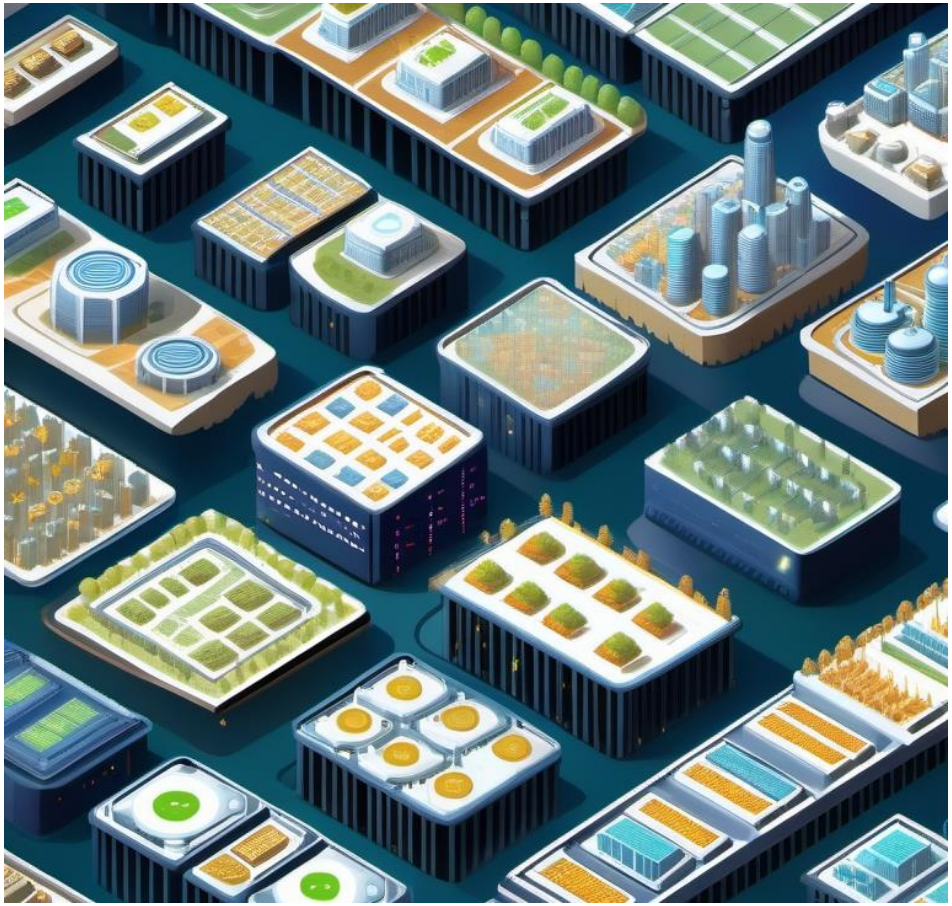
- Semantic Interoperability
  - Data formats
  - Knowledge definition
  - Data validation
- Data Catalogues
  - Data discovery
  - Meta description
- Policy-based privacy
- Identity providers
- Linked data services

ODRL Information Model 2.2
W3C Recommendation 15 February 2018

Decentralized Identifiers (DIDs) v1.0
Core architecture, data model, and representations

Linked Data Platform 1.0
W3C Recommendation 26 February 2015

INDEX



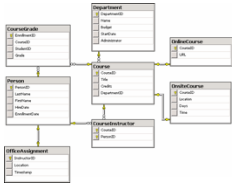
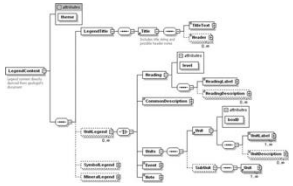
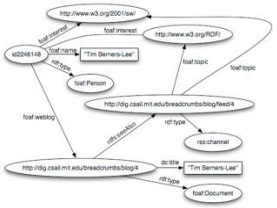
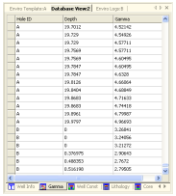
01. Introduction

02. RDF

03. Schemas and Ontologies

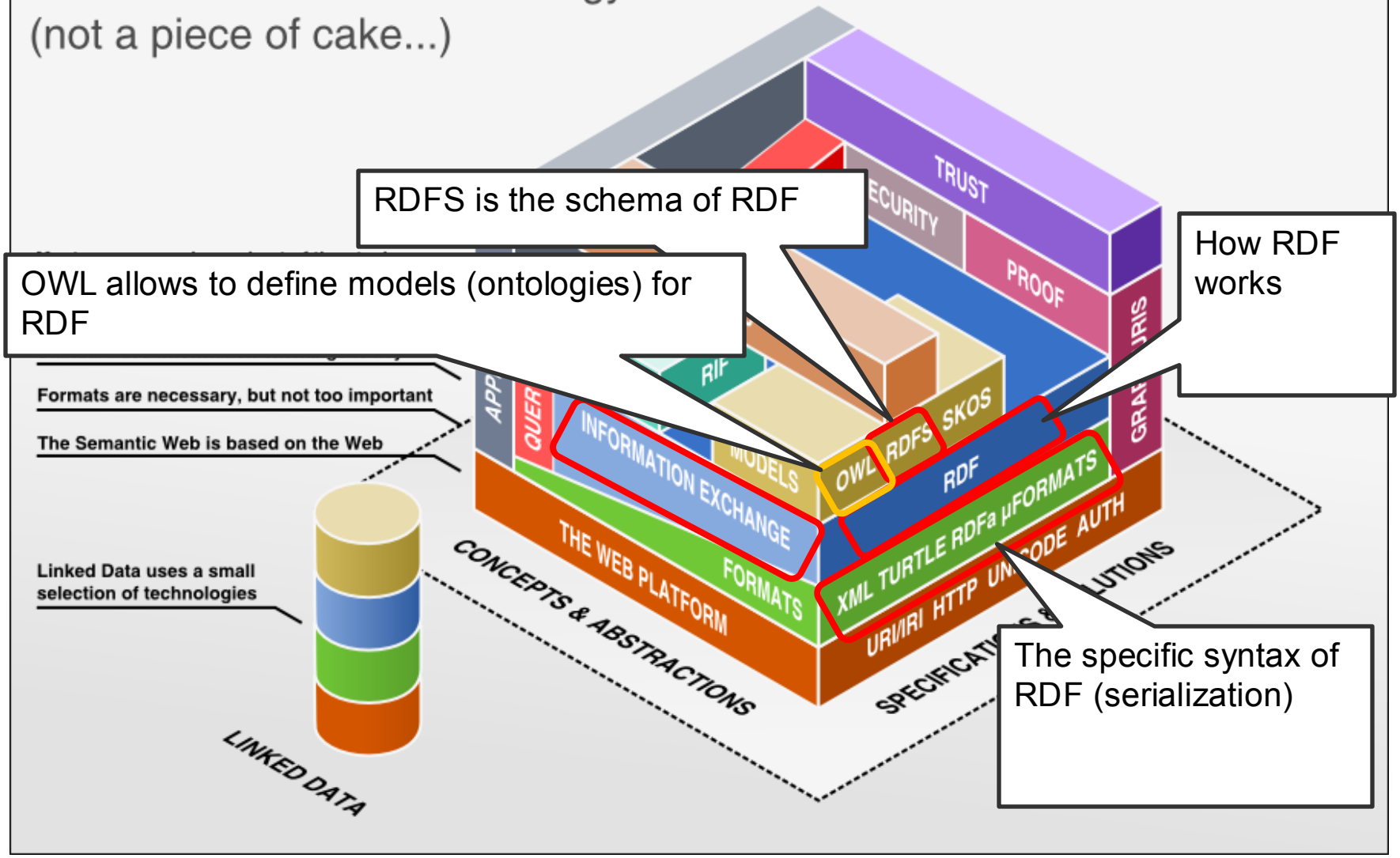
04. Sample cases

Resource Description Framework

	Database	XML	RDF(S)	
Schema				RDF Schema
Data				RDF

- W3C Recommendation
  - Model
  - Syntax
  - Semantics

The Semantic Web Technology Stack (not a piece of cake...)



Resource Description Framework

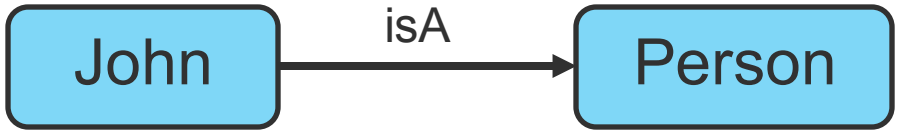
- Triplet format: subject predicate object
- RDF always describes a graph

Subject	- It can only be a URI or a blank node, - It is the identifier of the data associated with a real-world entity.
Predicate	- It can only be a URI, which must belong to an existing vocabulary/ontology.
Object	- It can be a URI, which must belong to an existing vocabulary/ontology and have been defined as a class. - Can be a URI or blank node, which may itself be a subject of another set of triples - Can be a string, string with type, or a primitive type (literal)

RDF

Triplet: subject predicate object

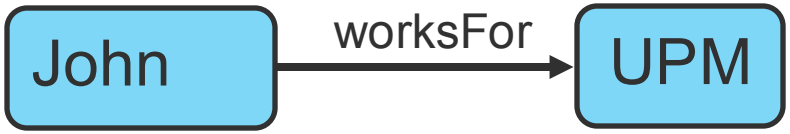
- John is a person



- John is named “John Doe”



- John works for a University

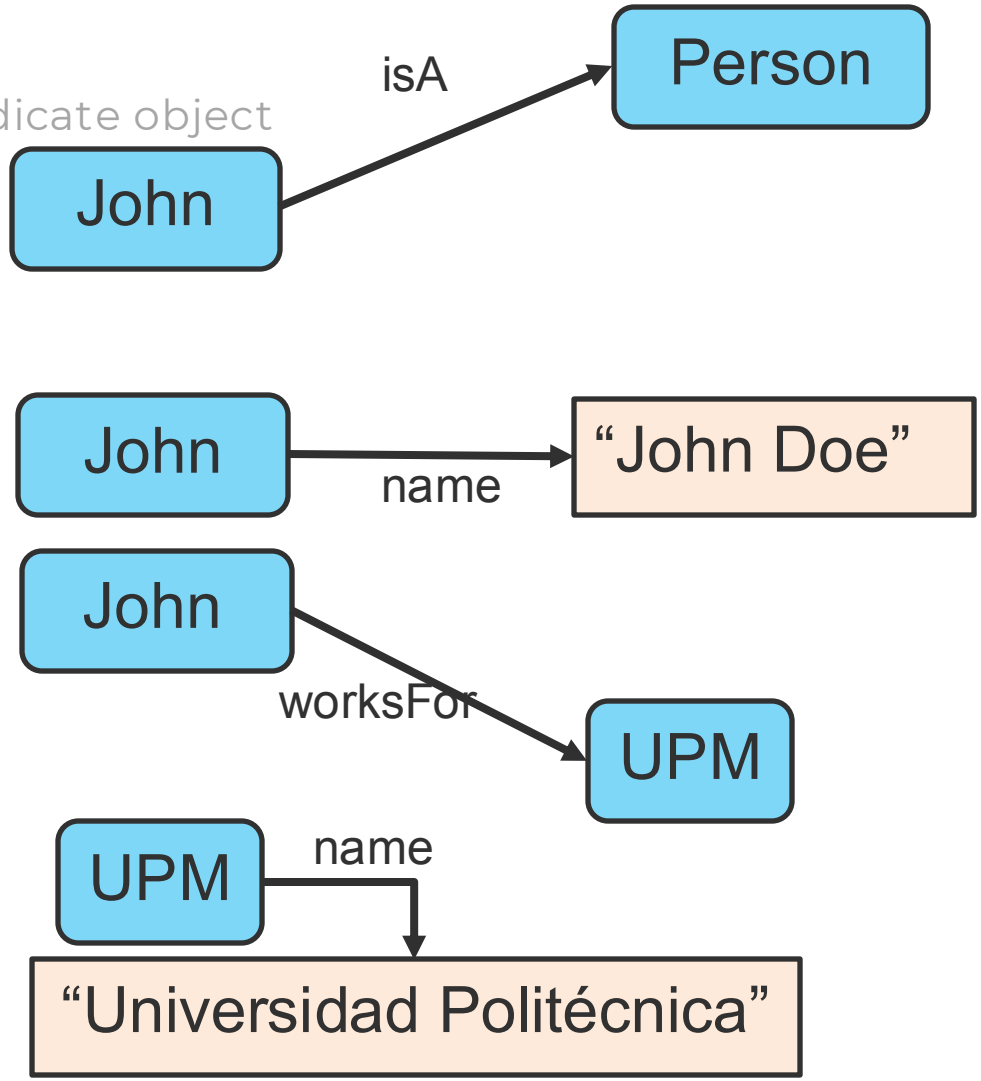


- The university is named Universidad Politécnica



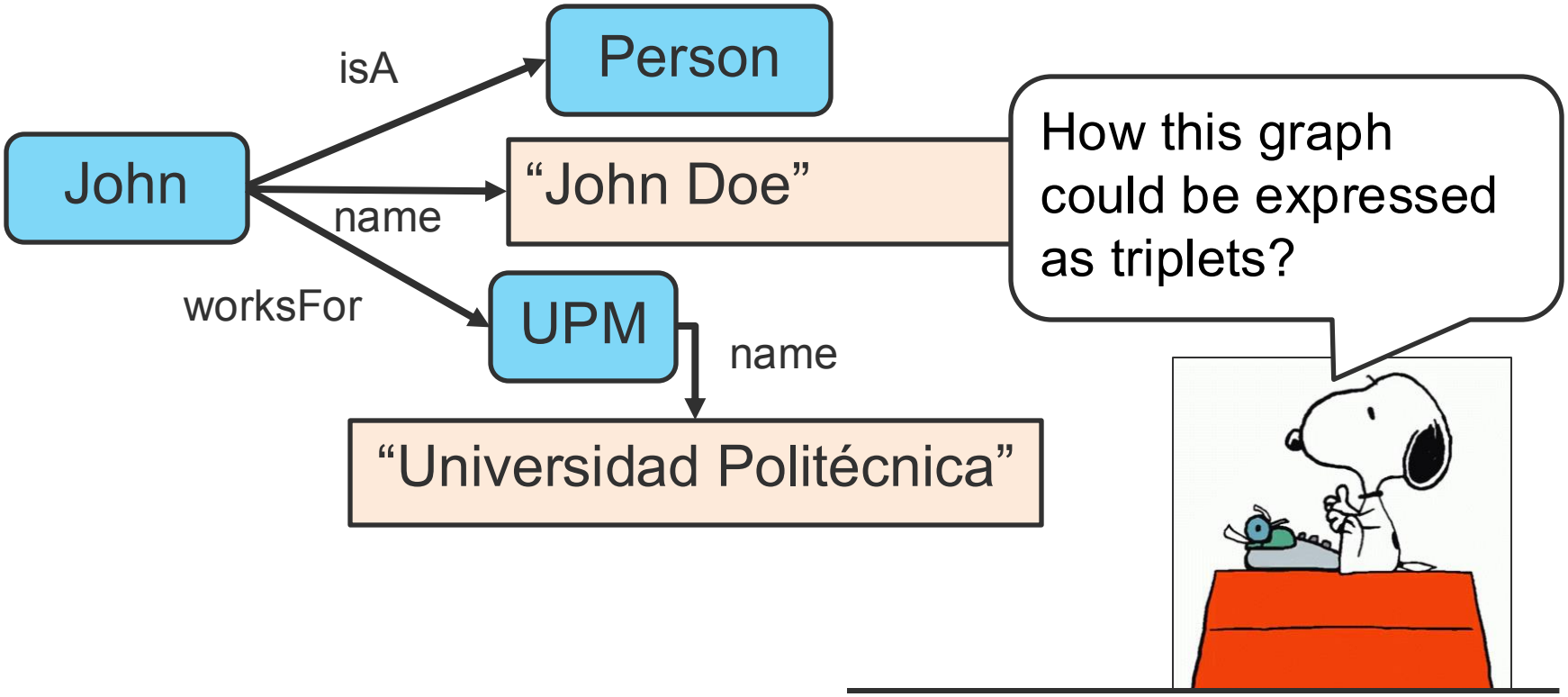
RDF

Triplet: subject predicate object

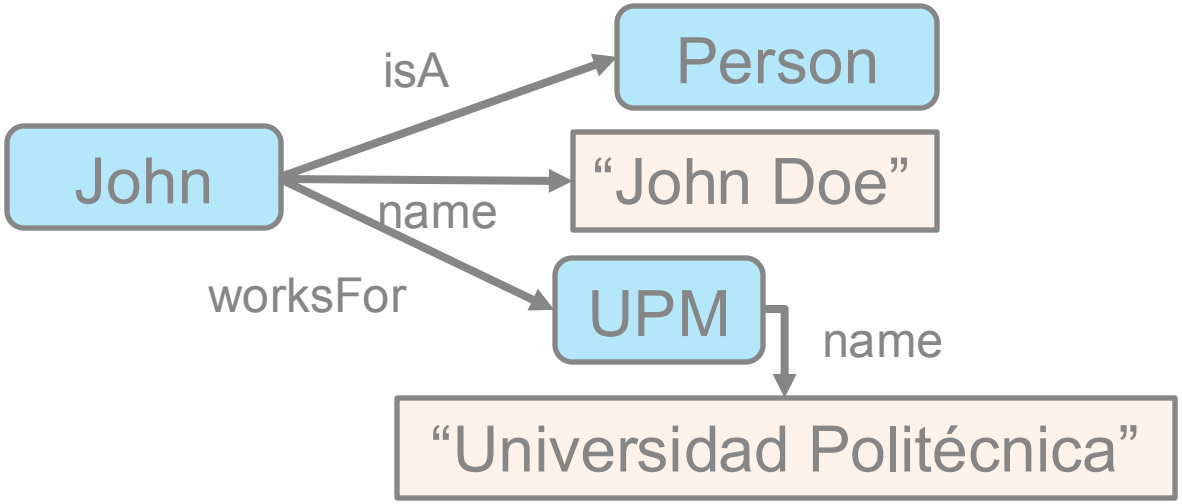


RDF

Triplet: subject predicate object



Subject	Predicate	Object
John	isA	Person
John	name	“John Doe”
John	worksFor	UPM
UPM	name	“Universidad Politécnica”



RDF

Triplet: subject predicate object

- Subject:
  - The identifier of the data associated with a real-world entity.
  - It can only be a **URI** or a **blank node.**

Suject	Predicate	Object
John	isA	Person
John	name	“John Doe”
John	worksFor	UPM
UPM	name	“Universidad Politécnica”

RDF

Triplet: subject predicate object

- Subject:
  - The identifier of the data associated with a real-world entity.
  - It can only be a **URI** or a **blank node.**

Suject	Predicate	Object
~~John~~	isA	Person
~~John~~	name	“John Doe”
~~John~~	worksFor	UPM
~~UPM~~	name	“Universidad Politécnica”

RDF

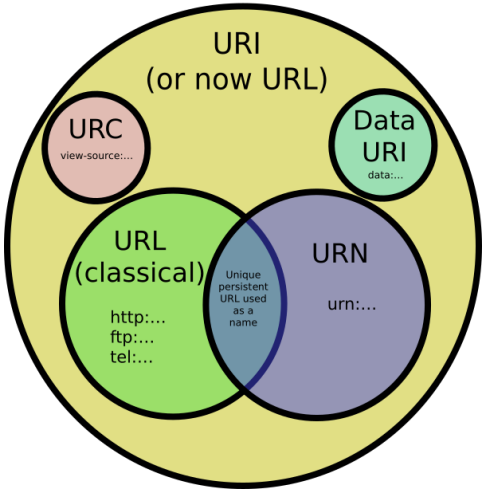
Triplet: subject predicate object

http://www.upm.es:8080/etsisi/personal?type=pid#John

Protocol | Domain | Route (path) | Filter | Local ID (fragment)

- IRI: International Resource Identifier
  - URI: Uniform Resource identifier
    - URL: Uniform Resource Locator
    - URN: Uniform Resource Names
      - urn:ISSN:0167-6423
      - urn:isbn:096139210x
    - UUID:
      - urn:uuid:3b91c1d8-...

Venn diagram of URIs as defined by the W3C



RDF

Triplet: **subject** predicate object

- Subject:
  - The identifier of the data associated with a real-world entity.
  - It can only be a **URI** or a **blank node.**

    http://www.upm.es/etsisi/resource/John

    http://www.upm.es/etsisi/resource/Escuela

Subject	Predicate	Object
http://www.upm.es/etsisi/resource/John	isA	Person
http://www.upm.es/etsisi/resource/John	name	“John Doe”
http://www.upm.es/etsisi/resource/John	worksFor	UPM
http://www.upm.es/etsisi/resource/Escuela	name	“Universidad Politécnica”

RDF

Triplet: subject predicate object

- The *namespace* is all those terms that hang on the common domain.
- The common domain can be "prefixed" to facilitate RDF writing.
  - For instance, instead of http://xmlns.com/foaf/0.1/name it be can used **foaf:name**



RDF

Triplet: **subject** predicate object

- Subject:
  - The identifier of the data associated with a real-world entity.
  - It can only be a **URI** or a **blank node.**

Prefix dat: http://www.upm.es/etsisi/resource/

Subject	Predicate	Object
dat:John	isA	Person
dat:John	name	“John Doe”
dat:John	worksFor	UPM
dat:Escuela	name	“Universidad Politécnica”

RDF

Triplet: subject predicate object

- Name space syntax:

http://www.upm.es:8080/etsisi/personal?type=pid#John

http://www.upm.es/etsisi/resource/





RDF

Triplet: subject **predicate** object

- Predicate:
  - It can only be a **URI**, which **must belong to an existing vocabulary/ontology.**
  - Let's use for example the FOAF vocabulary: http://xmlns.com/foaf/0.1/

Subject	Predicate	Object
dat:John	isA	Person
dat:John	name	"John Doe"
dat:John	worksFor	UPM
dat:Escuela	name	"Universidad Politécnica"
Prefixes		
dat: http://www.upm.es/etsisi/resources/		

RDF

Triplet: subject **predicate** object

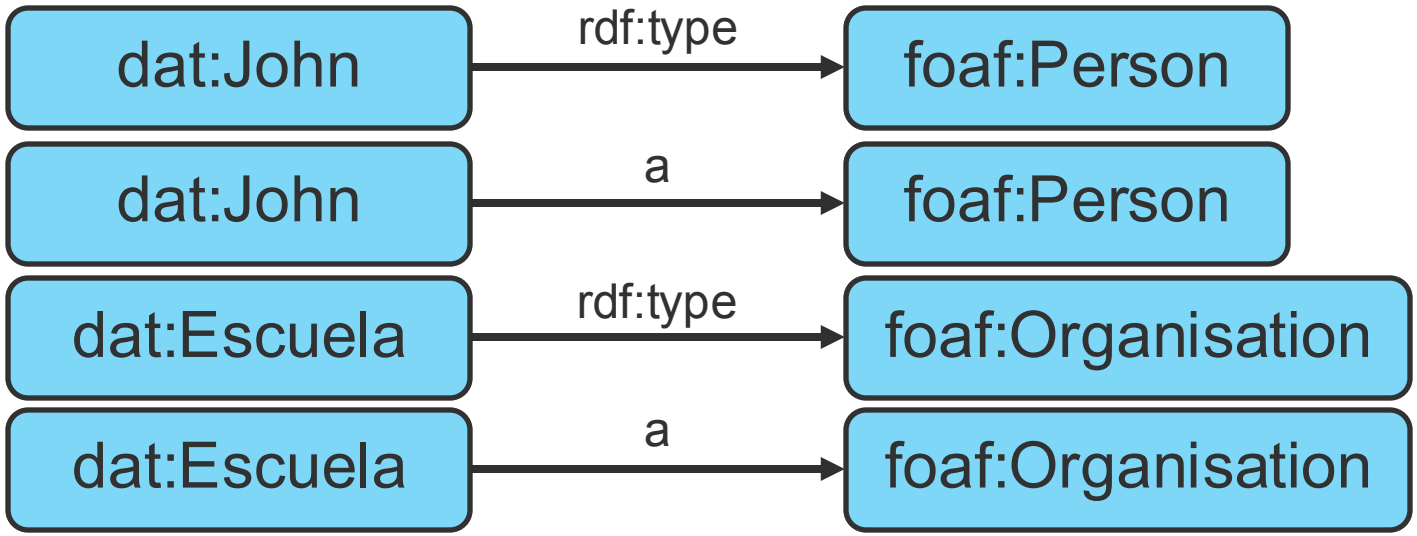
- Predicate:
  - It can only be a **URI**, which **must belong to an existing vocabulary/ontology.**
  - Let's use for example the FOAF vocabulary: http://xmlns.com/foaf/0.1/

Subject	Predicate	Object
dat:John	isA	Person
dat:John	**foaf:name**	"John Doe"
dat:John	**foaf:member**	UPM
dat:Escuela	**foaf:name**	"Universidad Politécnica"
Prefixes		
dat: http://www.upm.es/etsisi/resources/		
foaf: http://xmlns.com/foaf/0.1/		

RDF

Triplet: subject **predicate** object

- Any subject can be assigned a variable number (0 or infinity) of types.
- To assign a type is used as a predicate:
  - http://www.w3.org/1999/02/22-rdf-syntax-ns#type
  - rdf:type with prefix
  - The term 'a'
  - Subject, rdf:type, URI of the class



RDF

Triplet: subject **predicate** object

- Predicate:
  - It can only be a **URI**, which **must belong to an existing vocabulary/ontology.**

Subject	Predicate	Object
dat:John	**rdf:type**	Person
dat:John	foaf:name	“John Doe”
dat:John	foaf:member	UPM
dat:Escuela	foaf:name	“Universidad Politécnica”
Prefijos		
dat: http://www.upm.es/etsisi/resources/		
foaf: http://xmlns.com/foaf/0.1/		
rdf: http://www.w3.org/1999/02/22-rdf-syntax-ns#		

RDF

Triplet: subject predicate **object**

- Object:
  - It can be a **URI**, which **must belong to an existing vocabulary/ontology** and **have been defined as a class (type, object properties).**
  - Can be **a string with type**, or a **primitive type (data properties)**
  - Can be a **URI** or **blank node**, which may itself **be a subject of another set of triples (relation, object properties)**

Subject	Predicate	Object
dat:John	rdf:type	Person
dat:John	foaf:name	“John Doe”
dat:John	foaf:member	UPM
dat:Escuela	foaf:name	“Universidad Politécnica”

Prefijos
dat: http://www.upm.es/etsisi/resources/
foaf: http://xmlns.com/foaf/0.1/
rdf: http://www.w3.org/1999/02/22-rdf-syntax-ns#

RDF

Triplet: subject predicate **object**

- Object:
  - It can be a **URI**, which **must belong to an existing vocabulary/ontology** and **have been defined as a class (type, object properties).**

Subject	Predicate	Object
dat:John	**rdf:type**	**foaf:Person**
dat:John	foaf:name	"John Doe"
dat:John	foaf:member	UPM
dat:Escuela	foaf:name	"Universidad Politécnica"

Prefijos
dat: http://www.upm.es/etsisi/resources/
foaf: http://xmlns.com/foaf/0.1/
rdf: http://www.w3.org/1999/02/22-rdf-syntax-ns#

RDF

Triplet: subject predicate **object**

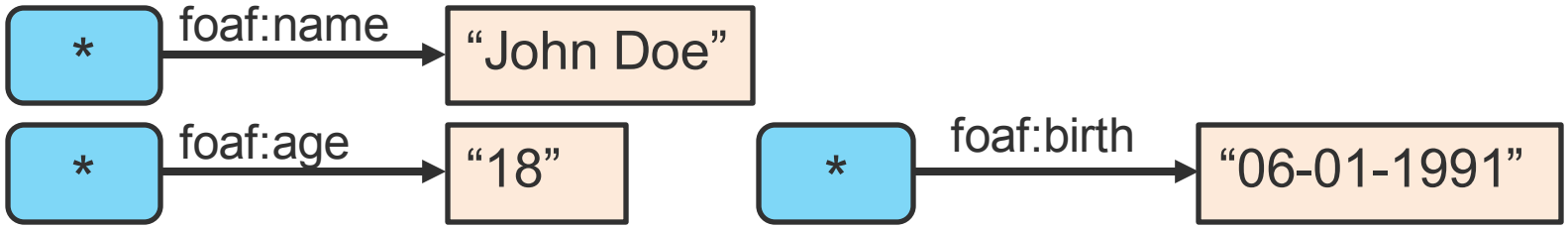
- Object:
  - Can be **a string with type**, or a **primitive type (data properties)**

Subject	Predicate	Object
dat:John	rdf:type	foaf:Person
dat:John	**foaf:name**	"John Doe"
dat:John	foaf:member	UPM
dat:Escuela	**foaf:name**	"Universidad Politécnica"
Prefijos		
dat: http://www.upm.es/etsisi/resources/		
foaf: http://xmlns.com/foaf/0.1/		
rdf: http://www.w3.org/1999/02/22-rdf-syntax-ns#		

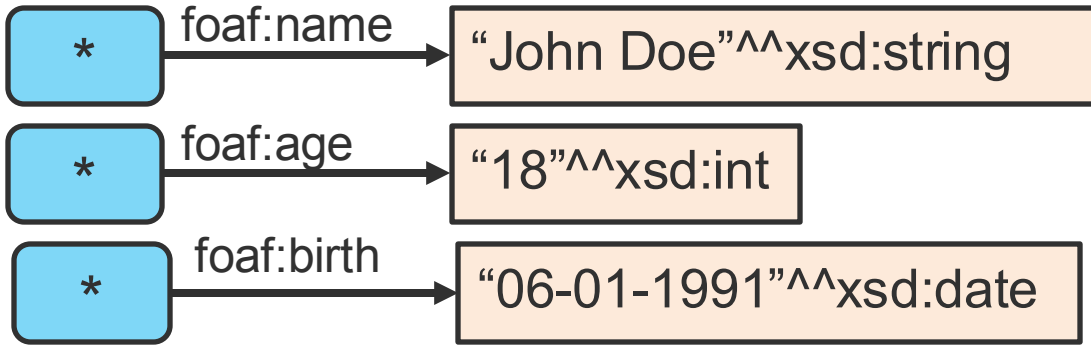
RDF (data properties)

Triplet: subject predicate **object**

- **They are predicates whose range is Literal**
  - String



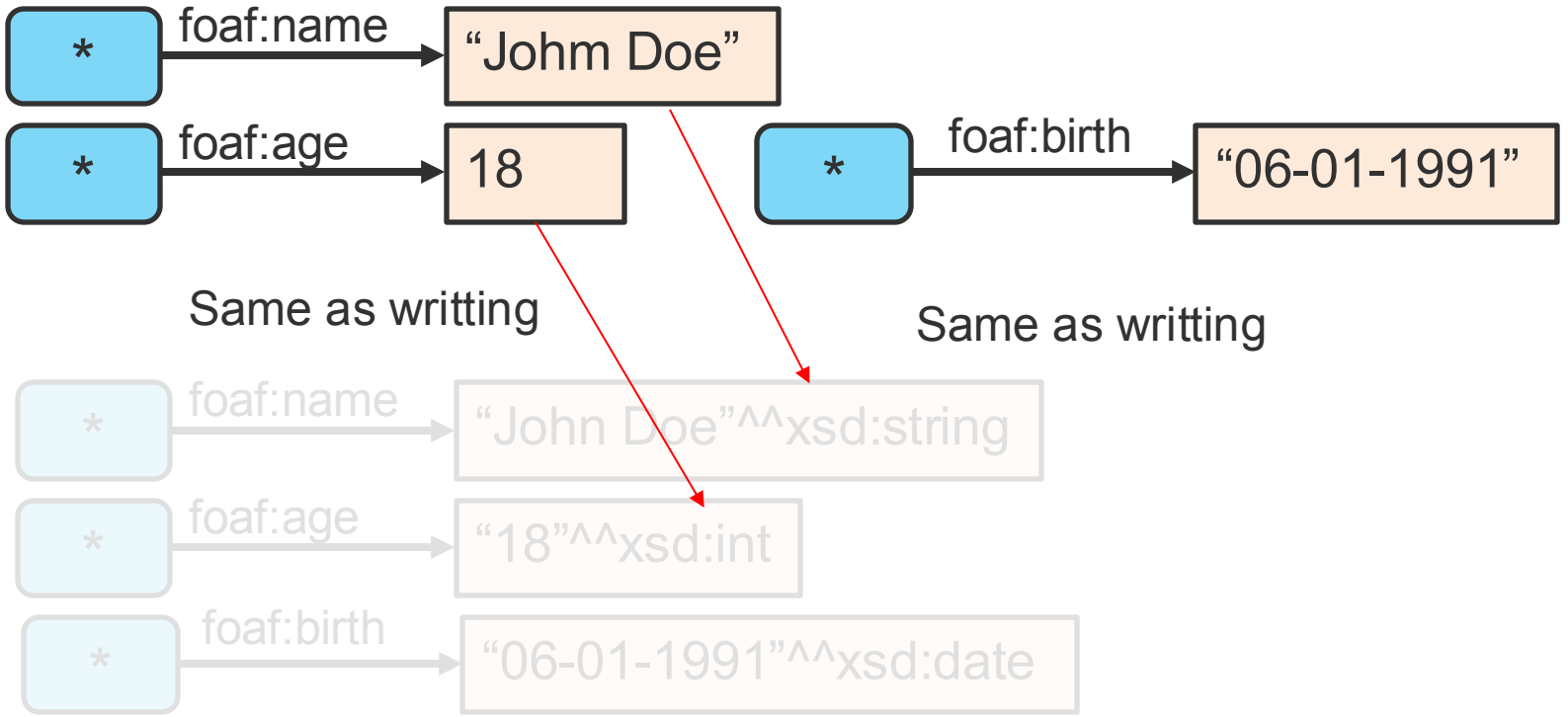
  - String with type



RDF (data properties)

Triplet: subject predicate **object**

- **They are predicates whose range is Literal**
  - Primitive type



RDF (data properties)

Triplet: subject predicate **object**

A list of the RDF-compatible XSD types, with short descriptions"

	Datatype	Value space (informative)
Core types	xsd:string	Character strings (but not all Unicode character strings)
	xsd:boolean	true, false
	xsd:decimal	Arbitrary-precision decimal numbers
	xsd:integer	Arbitrary-size integer numbers
IEEE floating-point numbers	xsd:double	64-bit floating point numbers incl. ±Inf, ±0, NaN
	xsd:float	32-bit floating point numbers incl. ±Inf, ±0, NaN
Time and date	xsd:date	Dates (yyyy-mm-dd) with or without timezone
	xsd:time	Times (hh:mm:ss.sss…) with or without timezone
	xsd:dateTime	Date and time with or without timezone
	xsd:dateTimeStamp	Date and time with required timezone
Recurring and partial dates	xsd:gYear	Gregorian calendar year
	xsd:gMonth	Gregorian calendar month
	xsd:gDay	Gregorian calendar day of the month
	xsd:gYearMonth	Gregorian calendar year and month
	xsd:gMonthDay	Gregorian calendar month and day
	xsd:duration	Duration of time
	xsd:yearMonthDuration	Duration of time (months and years only)
	xsd:dayTimeDuration	Duration of time (days, hours, minutes, seconds only)

RDF (data properties)

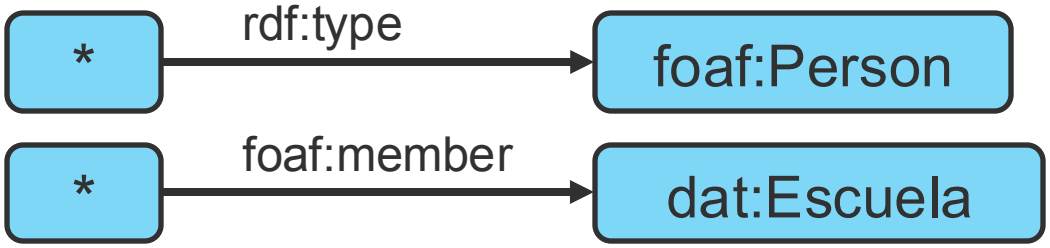
Triplet: subject predicate **object**

Limited-range integer numbers	xsd:byte	-128…+127 (8 bit)
	xsd:short	-32768…+32767 (16 bit)
	xsd:int	-2147483648…+2147483647 (32 bit)
	xsd:long	-9223372036854775808…+9223372036854775807 (64 bit)
	xsd:unsignedByte	0…255 (8 bit)
	xsd:unsignedShort	0…65535 (16 bit)
	xsd:unsignedInt	0…4294967295 (32 bit)
	xsd:unsignedLong	0…18446744073709551615 (64 bit)
	xsd:positiveInteger	Integer numbers >0
	xsd:nonNegativeInteger	Integer numbers ≥0
	xsd:negativeInteger	Integer numbers <0
	xsd:nonPositiveInteger	Integer numbers ≤0
Encoded binary data	xsd:hexBinary	Hex-encoded binary data
	xsd:base64Binary	Base64-encoded binary data
Miscellaneous XSD types	xsd:anyURI	Absolute or relative URIs and IRIs
	xsd:language	Language tags per [BCP47]
	xsd:normalizedString	Whitespace-normalized strings
	xsd:token	Tokenized strings
	xsd:NMTOKEN	XML NMTOKENs
	xsd:Name	XML Names
	xsd:NCName	XML NCNames

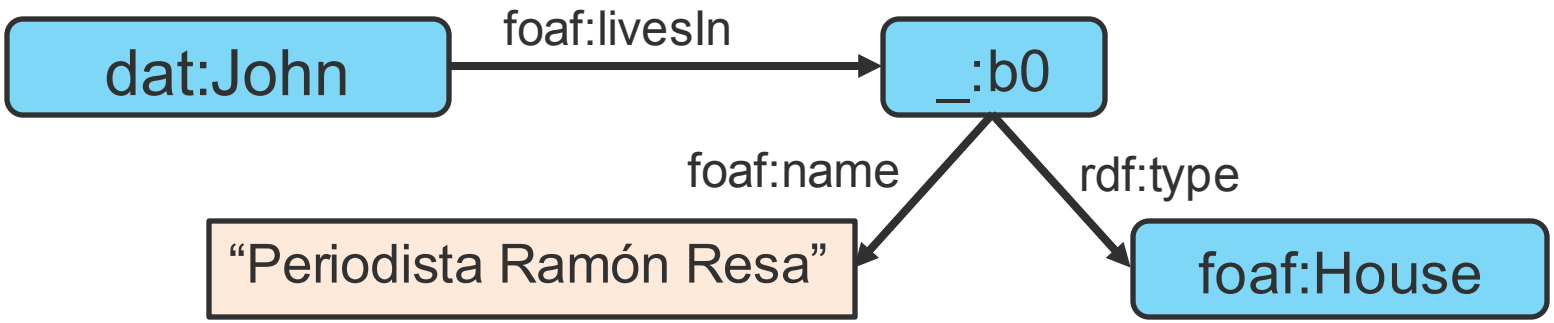
RDF (object properties)

Triplet: subject predicate **object**

- **Predicate whose range is a URI or blank node.**



  - Blank nodes are subjects that lack a unique identifier to be retrieved by themselves.



RDF

Triplet: subject predicate **object**

- Object:
  - Can be a **URI** or **blank node**, which may itself **be a subject of another set of triples (relation, object properties)**

Subject	Predicate	Object
dat:John	rdf:type	foaf:Person
dat:John	foaf:name	“John Doe”
dat:John	**foaf:member**	**dat:Escuela**
dat:Escuela	foaf:name	“Universidad Politécnica”

Prefijos
dat: http://www.upm.es/etsisi/resources/
foaf: http://xmlns.com/foaf/0.1/
rdf: http://www.w3.org/1999/02/22-rdf-syntax-ns#

RDF (summary)

Triplet: subject predicate object

- RDF is a graph expressed as a set of triplets
  - The URIs of the graph can be prefixed for improving readability
- Subjects
  - Identify a real-world entity
  - Must be URIs or Blank nodes
- Predicates
  - They have to be URIs defined in a vocabulary/ontology.
  - Data properties: if the range of the predicate is a literal
  - Object properties: if the range of the predicate is a URI or blank node
  - *rdf:type or a is the predicate that is used to assign a class to a subject.*
- Objects
  - They can be URIs (which can be subjects or classes of finite ontology), blank nodes (which are themselves subjects), or Literals (string, string with type, or primitive type).

RDF Serializations

- RDF serialisation is the way in which the triples (data syntax) are written.
- Many serialisations exist
  - **RDF/XML** (www.w3.org/TR/rdf-syntax-grammar/)
  - **Turtle** (http://www.w3.org/TR/turtle/)
  - **N-Triples** (http://www.w3.org/TR/n-triples/)
  - **TriG** (http://www.w3.org/TR/trig/)
  - **RDFa** (http://www.w3.org/TR/xhtml-rdfa-primer/)
  - **JSON-LD** (http://www.w3.org/TR/json-ld/)
  - **N-Quads** (http://www.w3.org/TR/n-quads/)
  - …
- Data in one serialisation can be transformed into another, and vice versa, without any inconvenience or loss.

Serializations: RDF/XML

```
<?xml version="1.0" encoding="utf-8" ?>
<rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
         xmlns:foaf="http://xmlns.com/foaf/0.1/">

  <foaf:Person rdf:about="http://www.upm.es/etsisi/resources/ John">
    <foaf:name> John Doe</foaf:name>
    <foaf:member>
      <rdf:Description rdf:about="http://www.upm.es/etsisi/resources/ Escuela">
        <foaf:name>Universidad Politécnica</foaf:name>
      </rdf:Description>
    </foaf:member>

  </foaf:Person>

</rdf:RDF>
```

Subject	Predicate	Object
dat:John	rdf:type	Person
dat:John	foaf:name	“John Doe”
dat:John	**foaf:member**	**dat:Escuela**
dat:Escuela	foaf:name	“Universidad Politécnica”

Prefijos
dat: http://www.upm.es/etsisi/resources/
foaf: http://xmlns.com/foaf/0.1/
rdf: http://www.w3.org/1999/02/22-rdf-syntax-ns#

Serializations : N-TRIPLES

<http://www.upm.es/etsisi/resources/John> <http://www.w3.org/1999/02/22-rdf-syntax-ns#type> <http://xmlns.com/foaf/0.1/Person> .
<http://www.upm.es/etsisi/resources/John> <http://xmlns.com/foaf/0.1/name> ”John Doe" .
<http://www.upm.es/etsisi/resources/John> <http://xmlns.com/foaf/0.1/member> <http://www.upm.es/etsisi/resources/Escuela> .
<http://www.upm.es/etsisi/resources/Escuela> <http://xmlns.com/foaf/0.1/name> "Universidad Politénica" .

LACK PREFIXES!

Subject	Predicate	Object
dat:John	rdf:type	Person
dat:John	foaf:name	“John Doe”
dat:John	**foaf:member**	**dat:Escuela**
dat:Escuela	foaf:name	“Universidad Politécnica”
Prefijos		
dat: http://www.upm.es/etsisi/resources/		
foaf: http://xmlns.com/foaf/0.1/		
rdf: http://www.w3.org/1999/02/22-rdf-syntax-ns#		

Serializations: N-TRIPLES

```
@prefix foaf: <http://xmlns.com/foaf/0.1/> .
@prefix dat: <http://www.upm.es/etsisi/resources/ > .

dat:John a foaf:Person ;
        foaf:name " John Doe" ;
        foaf:member dat:Escuela .

dat:Escuela foaf:name "Universidad Politécnica" .
```

Subject	Predicate	Object
dat:John	rdf:type	Person
dat:John	foaf:name	“John Doe”
dat:John	**foaf:member**	**dat:Escuela**
dat:Escuela	foaf:name	“Universidad Politécnica”
Prefijos		
dat: http://www.upm.es/etsisi/resources/		
foaf: http://xmlns.com/foaf/0.1/		
rdf: http://www.w3.org/1999/02/22-rdf-syntax-ns#		

Serializations: JSON-LD

```
[ {
   "@id": "http://www.upm.es/etsisi/resources/John",
   "@type": [ "http://xmlns.com/foaf/0.1/Person" ],
   "http://xmlns.com/foaf/0.1/name": [ { "@value": "John Doe" }],
   "http://xmlns.com/foaf/0.1/member": [ { "@id":
"http://www.upm.es/etsisi/resources/Escuela" }]
  }, {
   "@id": "http://www.upm.es/etsisi/ resources /Escuela",
   "http://xmlns.com/foaf/0.1/name": [ { "@value": "Universidad Politécnica"}]
  },{  "@id": "http://xmlns.com/foa
]
```

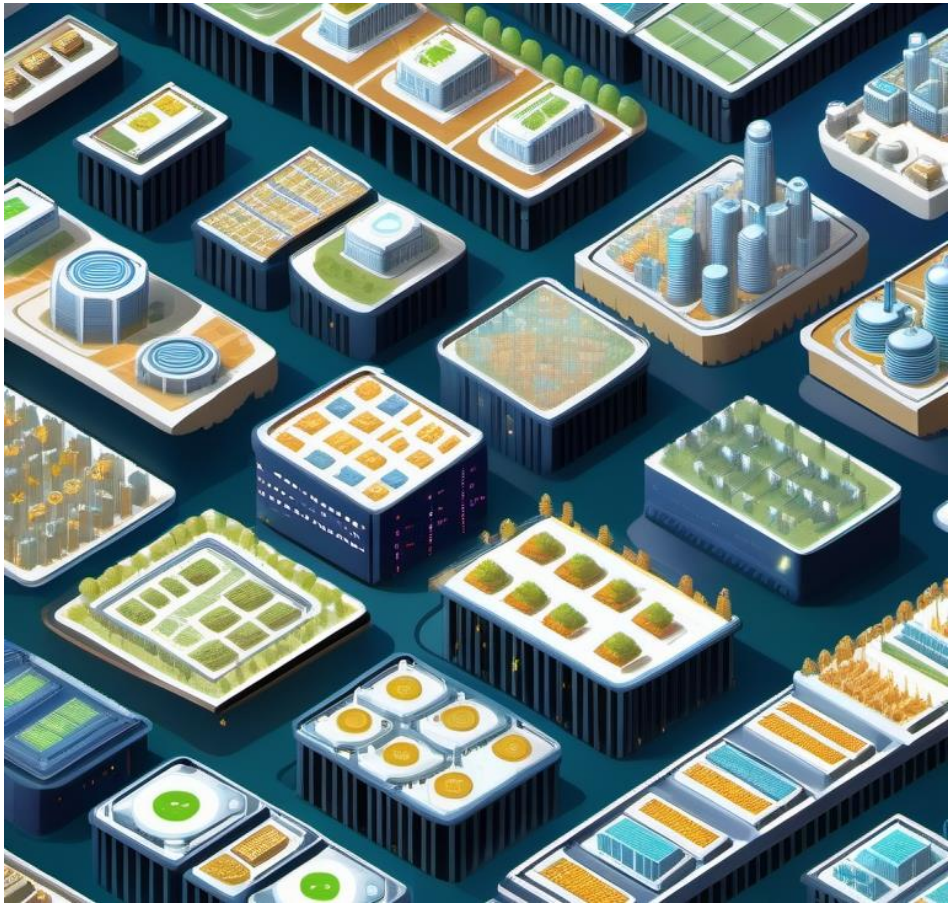
Subject	Predicate	Object
dat:John	rdf:type	Person
dat:John	foaf:name	“John Doe”
dat:John	**foaf:member**	**dat:Escuela**
dat:Escuela	foaf:name	“Universidad Politécnica”

Prefijos
dat: http://www.upm.es/etsisi/resources/
foaf: http://xmlns.com/foaf/0.1/
rdf: http://www.w3.org/1999/02/22-rdf-syntax-ns#

Serializations: JSON-LD 1.1

```
{
 "@context" : {
  "dat" : "http://www.upm.es/etsisi/resources/",
  "foaf" : "http://xmlns.com/foaf/0.1/",
  "Person" : "foaf:Person",
  "name" : "foaf:name",
  "worksFor" : "foaf:member"
  },
 "@id" : "dat:John",
 "@type" : "Person",
 "name"  : "John Doe",
 "worksFor" : {
      "@id" : "dat:Escuela",
   "name" : "Universidad Politécnica"
 }
}
```

Subject	Predicate	Object
dat:John	rdf:type	Person
dat:John	foaf:name	“John Doe”
dat:John	**foaf:member**	**dat:Escuela**
dat:Escuela	foaf:name	“Universidad Politécnica”
Prefijos		
dat: http://www.upm.es/etsisi/resources/		
foaf: http://xmlns.com/foaf/0.1/		
rdf: http://www.w3.org/1999/02/22-rdf-syntax-ns#		



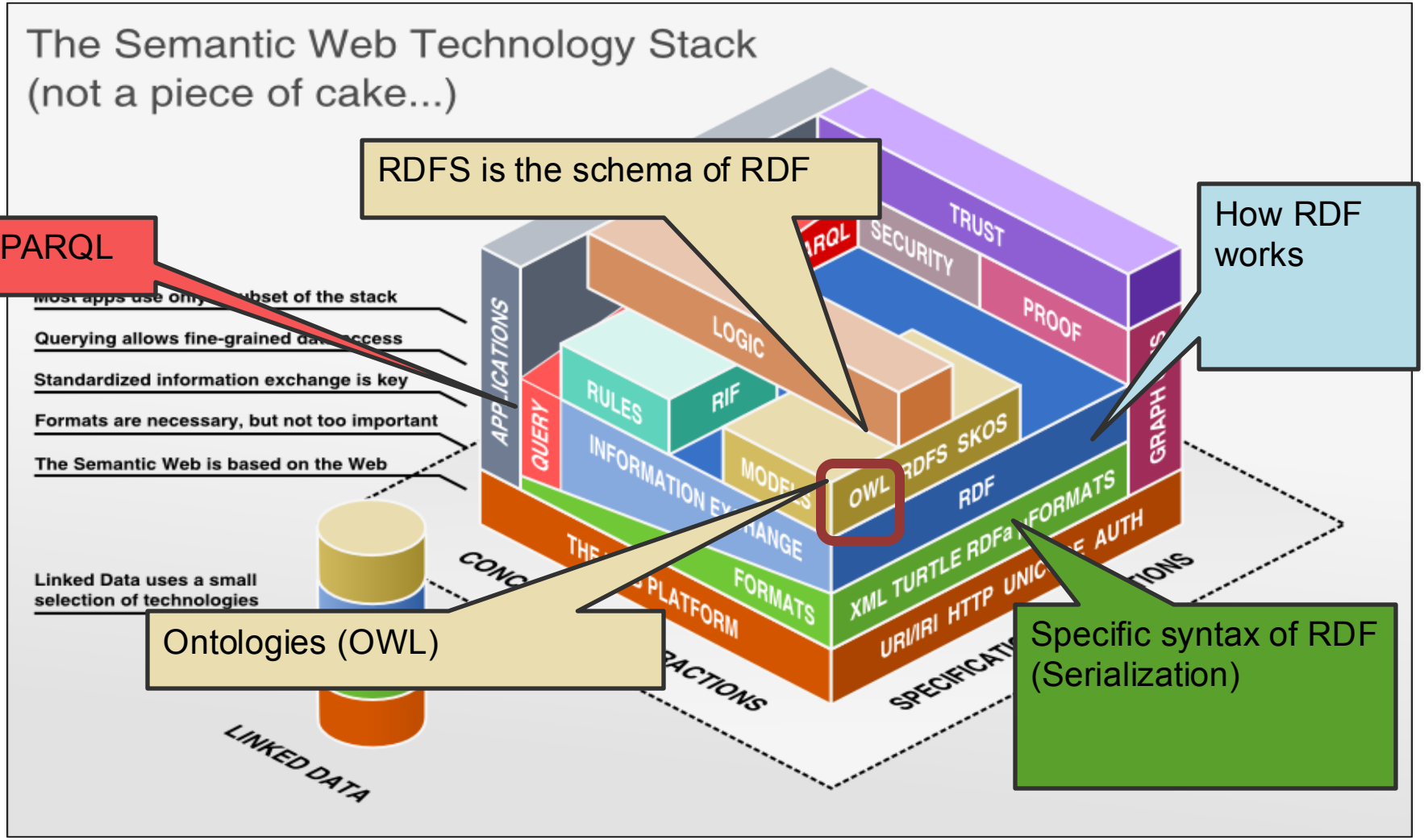
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03. Schemas and Ontologies

04. Sample cases

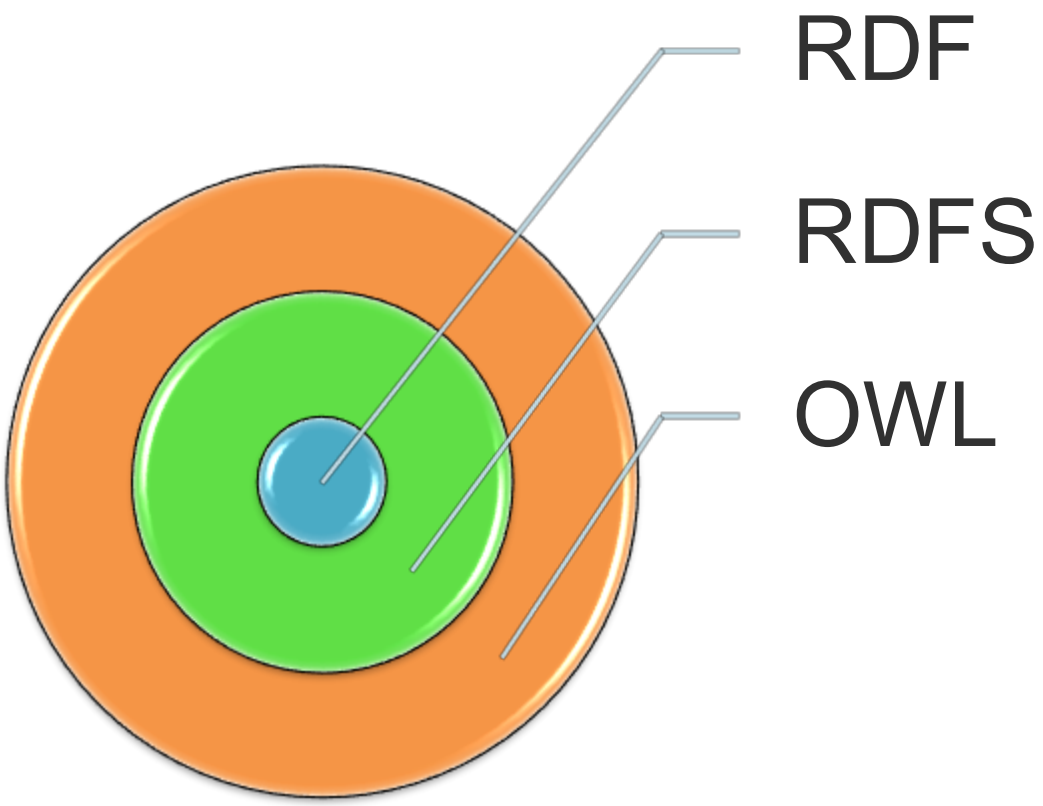

The Semantic Web Technology Stack
(not a piece of cake...)
RDFS is the schema of RDF
SPARQL
How RDF works
Ontologies (OWL)
Specific syntax of RDF (Serialization)
Querying allows fine-grained
Standardized information exchange is key
Formats are necessary, but not too important
The Semantic Web is based on the Web
Linked Data uses a small selection of technologies
TRUST
SECURITY
PROOF
LOGIC
RULES
RIF
OWL
RDF
SKOS
XML
LINKED DATA

Ontologies

Models expressed with the OWL language

- OWL stands for Web Ontology Language
- It is a W3C standard
  - https://www.w3.org/TR/owl2-overview/
- OWL has explicit formal semantics
  - Knowledge can be defined in a machine-interpretable form

Ontologies and schemes



Características del esquema RDF:

- Class (Thing, Nothing)
- rdfs:subClassOf
- rdf:Property
- rdfs:subPropertyOf
- rdfs:domain
- rdfs:range
- Individual

(Des)Igualdad:

- equivalentClass
- equivalentProperty
- sameAs
- differentFrom
- AllDifferent
- distinctMembers

Características de propiedad :

- ObjectProperty
- DatatypeProperty
- inverseOf
- TransitiveProperty
- SymmetricProperty
- FunctionalProperty
- InverseFunctionalProperty

Restricciones de propiedad:

- Restriction
- onProperty
- allValuesFrom
- someValuesFrom

Cardinalidad restringida:

- minCardinality (solo 0 or 1)
- maxCardinality (solo 0 or 1)
- cardinality (solo 0 or 1)

Información de cabecera:

- Ontology
- imports

Intersección de clases :

- intersectionOf

Tipos de datos:

- xsd datatypes

Control de versiones:

- versionInfo
- priorVersion
- backwardCompatibleWith
- incompatibleWith
- DeprecatedClass
- DeprecatedProperty

propiedades de anotación:

- rdfs:label
- rdfs:comment
- rdfs:seeAlso
- rdfs:isDefinedBy
- AnnotationProperty
- OntologyProperty

OWL summary

Ontologies

Ontology definition

```
@prefix : <http://www.example.org/vocab#>.
@prefix owl: <http://www.w3.org/2002/07/owl#> .
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix xml: <http://www.w3.org/XML/1998/namespace> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@base <http://www.linkeddata.es/clases/ontologia#> .

<http://www.example.org/vocab#> rdf:type owl:Ontology ;
                                rdfs:comment "Una ontologia como tal" ;
                                rdfs:label "Ontologia de prueba" ;
                                owl:versionInfo 1.0 .
```

Ontologies

Definition of classes and subclasses

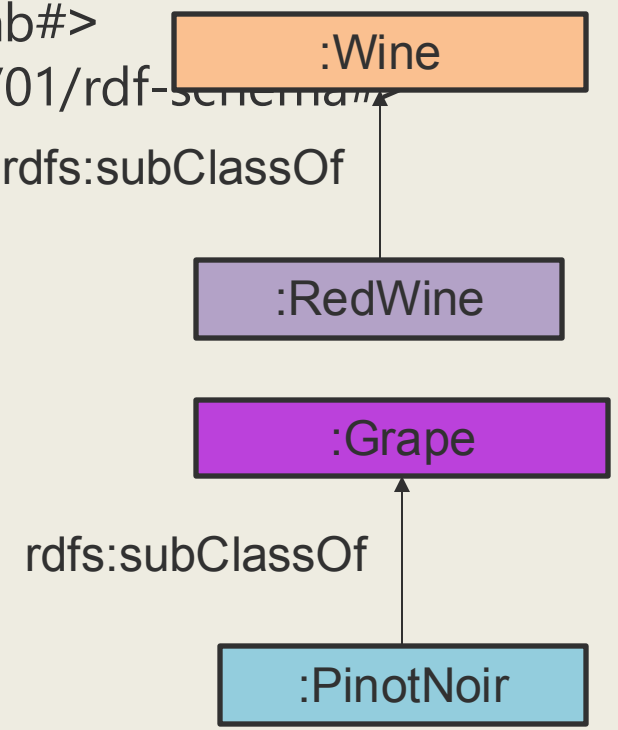
```
@prefix owl: <http://www.w3.org/2002/07/owl#>
@prefix  : <http://www.example.org/vocab#>
@prefix  rdfs : <http://www.w3.org/2000/01/rdf-schema#>

:Wine a owl:Class .

:RedWine a owl:Class ;
        rdfs:subClassOf :Wine .

:Grape a owl:Class ;

:PinotNoir a owl:Class ;
        rdfs:subClassOf :Grape .
```



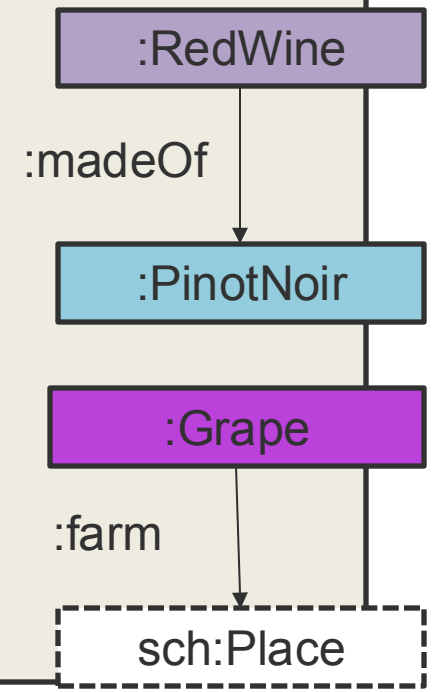
Ontologies

Definition of classes relationships (object properties)

```
@prefix owl: <http://www.w3.org/2002/07/owl#>
@prefix  : <http://www.example.org/vocab#>
@prefix  rdfs : <http://www.w3.org/2000/01/rdf-schema#>


:madeOf a owl:ObjectProperty ;
            rdfs:domain :RedWine ;
            rdfs:range    :PinotNoir .

:farm a owl:ObjectProperty ;
            rdfs:domain :Grape ;
            rdfs:range    :sch:Place .
```



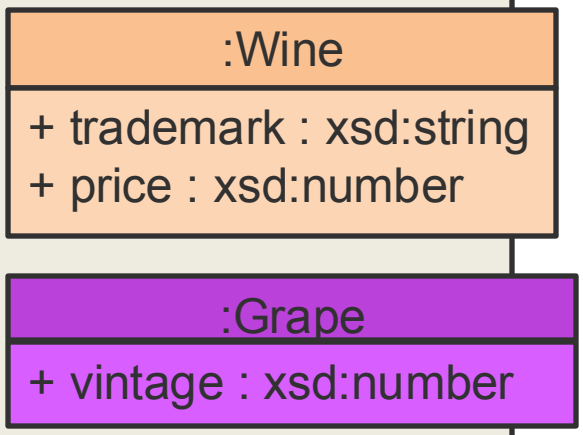
Ontologies

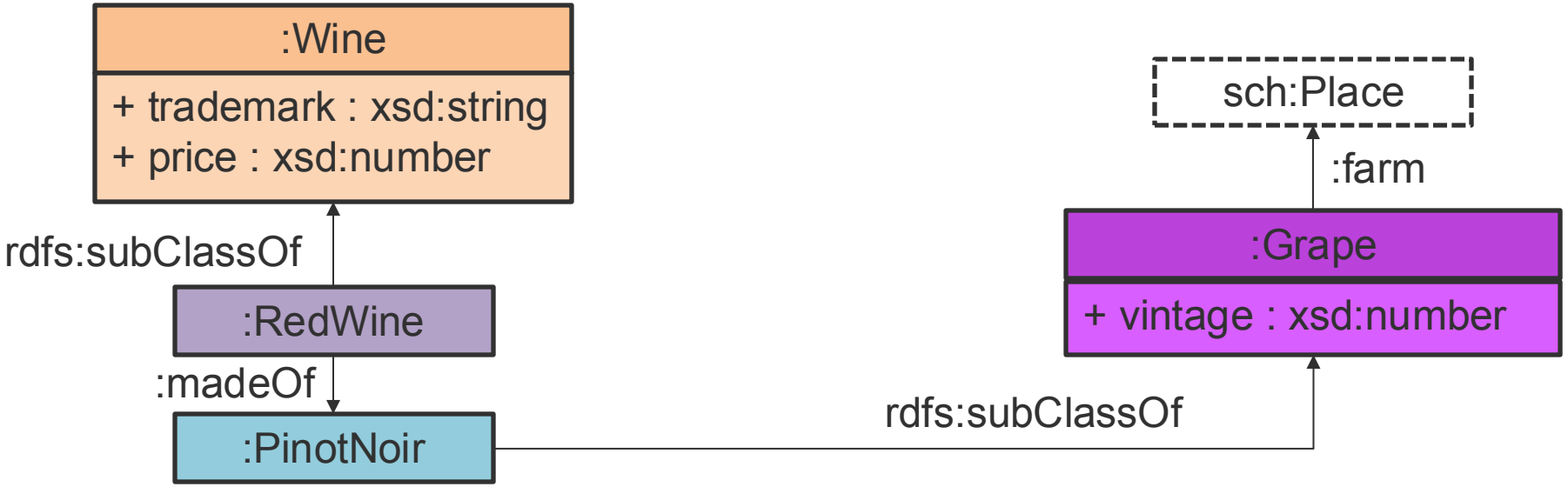
Attributes definition (data properties)

```
@prefix owl: <http://www.w3.org/2002/07/owl#>
@prefix  : <http://www.example.org/vocab#>
@prefix  rdfs : <http://www.w3.org/2000/01/rdf-schema#>
:tradeMark a owl:DatatypeProperty ;
            rdfs:domain :Wine ;
            rdfs:range    xsd:string .

:price a owl:DatatypeProperty ;
            rdfs:domain :RedWine ;
            rdfs:range    xsd:number .

:vintage a owl:DatatypeProperty ;
            rdfs:domain :Grape ;
            rdfs:range    xsd:integer .
```






```
ex:wine1 a :RedWine;
        :trademark “El pago de los capellanes” ;
        :price 34.3 ;
        :madeOf [
                a :PinotNoir
                :vintage “2018”
                :farm <https://es.dbpedia.org/page/Provincia_de_Burgos>
        ] .
```

Ontologies

Other relationships and terms

- owl:sameAs
  - Allows linking two instances representing the same real-world object.
- owl:equivalentClass, owl:equivalentProperty
  - They allow linking terms from one ontology (classes or properties) with the homonyms of another, allowing terms from both ontologies to be used interchangeably.
- And many more: transitive properties, cardinalities in the relations, …
  - https://www.w3.org/TR/owl-guide/#Contents

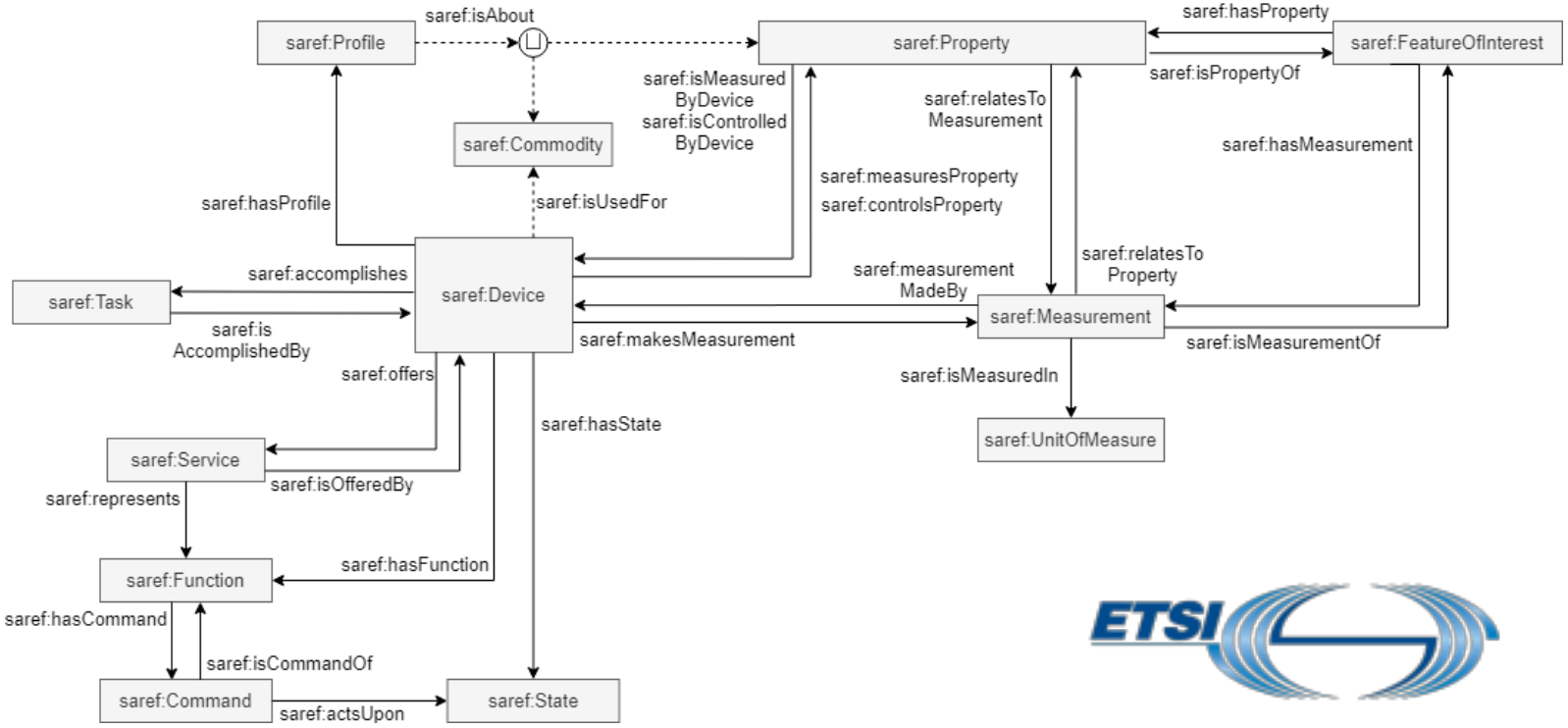
Ontologies vs Schemas

- Ontologies are built on top of schemas.
- Ontologies include additional information and constraints
- Reasoners work for both ontologies (owl) and RDF schema (rdfs)

- Known ontologies (https://lov.linkeddata.es/dataset/lov/):
  - Foaf (http://xmlns.com/foaf/0.1/)
  - Schema (https://schema.org/)
  - Geo (http://www.w3.org/2003/01/geo/wgs84_pos#)
  - ....

Sample standard ontology

Ontology for IoT: SAREF



Take away





Ontology development



Link

Resources

RDF

- Easy RDF

Ontologies

- Lov
- Schema
- Saref

Data

- Dbpedia
- Wikidata

Some research areas

RDF conceptualization

- RDF serializations
  - RDF *
  - Signing data
  - ....

Ontology engineering

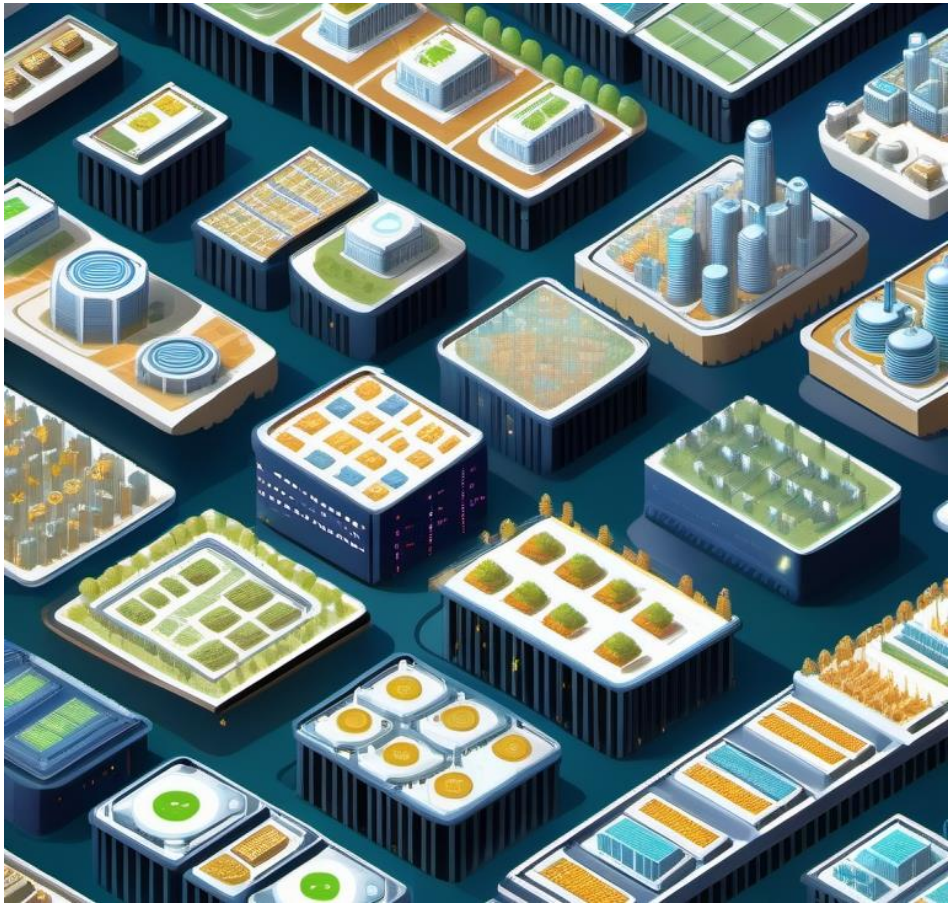
- Ontology development
  - LLMs
  - Visual languages
  - Engineering-based Methodologies
  - ....

Semantic interoperability

- Ontologies/RDF alignments (OAEI)
  - 1 vs 1
  - Dynamic alignments
  - Blocking-based comparison
  - …



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03. Schemas and Ontologies

04. Sample cases

Case 1

Writing RDF mock RDF using the ontology of Biblioteca Nacional Española

- 1.1 Write an instance that is a copy and has a barcode, date of creation, loan and return, and price. In addition, it must indicate that the copy is in the Goya Room (SALA_GOYA).

- 1.2 Write a Resource instance that has an ISBN and a title (it has a title itself). Indicate that this resource has a copy which is the one defined above.

Case 1 - 1.1

Writing RDF mock RDF using the ontology of Biblioteca Nacional Española

```
@prefix dat: <http://example.org/data/> .
@prefix bne: <http://datos.bne.es/def/> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .

dat:Documento1 a bne:C1004;
    bne:P4005 "A-0050-Z";                                  # Codigo de barras
    bne:P4003 "2023-10-18+18:00"^^xsd:dateTime; # Fecha ultimo prestamo
    bne:P4004 "2023-10-19+10:00"^^xsd:dateTime; # Fecha ultima devolucion
    bne:P4015 "2023-10-18+06:00"^^xsd:dateTime; # Fecha de creacion
    bne:P4007 "SALA_GOYA" .                                # Localizacion
```

Case 1 - 1.2

Writing RDF mock RDF using the ontology of Biblioteca Nacional Española

```
@prefix dat: <http://example.org/data/> .
@prefix bne: <http://datos.bne.es/def/> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .

dat:Documento1 a bne:C1004;
   bne:P4005 "A-0050-Z";                              # Codigo de barras
   bne:P4003 "2023-10-18+18:00"^^xsd:dateTime; # Fecha ultimo prestamo
   bne:P4004 "2023-10-19+10:00"^^xsd:dateTime; # Fecha ultima devolucion
   bne:P4015 "2023-10-18+06:00"^^xsd:dateTime; # Fecha de creacion
   bne:P4007 "SALA_GOYA" .                            # Localizacion

dat:Recurso a bne:C1004 ;
   bne:P3013 "0-4362-1509-8” ;                        # ISBN
   bne:P3002 ”El mago de Oz” .                        # Titulo

dat:Recurso bne:OP3001 dat:Documento1 .        # Está ejemplificado por
```

Case 2

Writing RDF mock RDF using the ontology of Schema

- *2.1 Write an instance of an actor with a first name, surname and age.*
- *2.2 Write an instance of a television series with a title and a year. The title must be in English and Spanish.*
- *2.3 Relate the two instances.*

Case 2 – 2.1

Writing RDF mock RDF using the ontology of Schema

```
@prefix dat: <http://example.org/data/> .
@prefix sch: <https://schema.org/> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
@prefix owl: <http://www.w3.org/2002/07/owl#>.
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .

dat:Actor1 a sch:Person ;
   sch:name “Antonio Banderas” ; sch:givenName “Antonio ; sch:familyName “Banderas” ;
   voc:age 63 .

voc:age a owl:DatatypeProperty ;
       rdfs:domain sch:Person ;
        rdfs:range xsd:nonNegativeInteger .
```

Case 2 – 2.2

Writing RDF mock RDF using the ontology of Schema

```
@prefix dat: <http://example.org/data/> .
@prefix sch: <https://schema.org/> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
@prefix owl: <http://www.w3.org/2002/07/owl#>.
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .

dat:Actor1 a sch:Person ;
   sch:name “Antonio Banderas” ; sch:givenName “Antonio ; sch:familyName “Banderas” ;
   voc:age 63 .

voc:age a owl:DatatypeProperty ;
        rdfs:domain sch:Person ;
       rdfs:range xsd:nonNegativeInteger .

dat:Show a sch:TVShow;
  sch:name “Las aventuras del Zorro”@es, “Zorro the adventures”@en .
  sch:startDate  “2023-10-24T10:52:07Z” ;
  sch:endDate “2023-10-24T10:52:07Z” .
```

Case 2 – 2.3

Writing RDF mock RDF using the ontology of Schema

```
@prefix dat: <http://example.org/data/> .
@prefix sch: <https://schema.org/> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
@prefix owl: <http://www.w3.org/2002/07/owl#>.
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .

dat:Actor1 a sch:Person ;
   sch:name “Antonio Banderas” ; sch:givenName “Antonio ; sch:familyName “Banderas” ;
   voc:age 63 .

voc:age a owl:DatatypeProperty ;
         rdfs:domain sch:Person ;
       rdfs:range xsd:nonNegativeInteger .

dat:Show a sch:TVShow;
  sch:name “Las aventuras del Zorro”@es, “Zorro the adventures”@en .
  sch:startDate  “2023-10-24T10:52:07Z” ;
  sch:endDate “2023-10-24T10:52:07Z” .

dat:Show sch:actor dat:Actor1 .
```

Case 3

Writing RDF mock RDF using the ontology of FOAF

- 3.1 Write two instances that represent two people and each have a name. Both persons must know each other.
- 3.2 Write an instance of Organisation.
- 3.3 Indicate that one of the persons works in the Organisation.

Case 3 – 3.1

Writing RDF mock RDF using the ontology of FOAF

```
@prefix dat: <http://example.org/data/> .
@prefix foaf: <http://xmlns.com/foaf/0.1/> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .

dat:Person1 a foaf:Person ;
        dat:Person1 foaf:name “Pedro”@es, “Peter”@en;
         dat:Person1 foaf:knowns dat:Person2 .

dat:Person2 a foaf:Person;
        dat:Person2 foaf:name “Pablo”@es, “Paul”@en;
         dat:Person2 foaf:knowns dat:Person21 .
```

Case 3 – 3.2

Writing RDF mock RDF using the ontology of FOAF

```
@prefix dat: <http://example.org/data/> .
@prefix foaf: <http://xmlns.com/foaf/0.1/> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .

dat:Person1 a foaf:Person ;
        foaf:name “Pedro”@es, “Peter”@en;
        foaf:knowns dat:Person2 .

dat:Person2 a foaf:Person;
        foaf:name “Pablo”@es, “Paul”@en;
        foaf:knowns dat:Person21 .

dat:Org1 a foaf:Organization;
        dat:name “Cantera de Piedradura” .
```

Case 3 – 3.3

Writing RDF mock RDF using the ontology of FOAF

```
@prefix dat: <http://example.org/data/> .
@prefix foaf: <http://xmlns.com/foaf/0.1/> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
@prefix owl: <http://www.w3.org/2002/07/owl#>.
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .

dat:Person1 a foaf:Person ;
        foaf:name “Pedro”@es, “Peter”@en;
        foaf:knowns dat:Person2 .

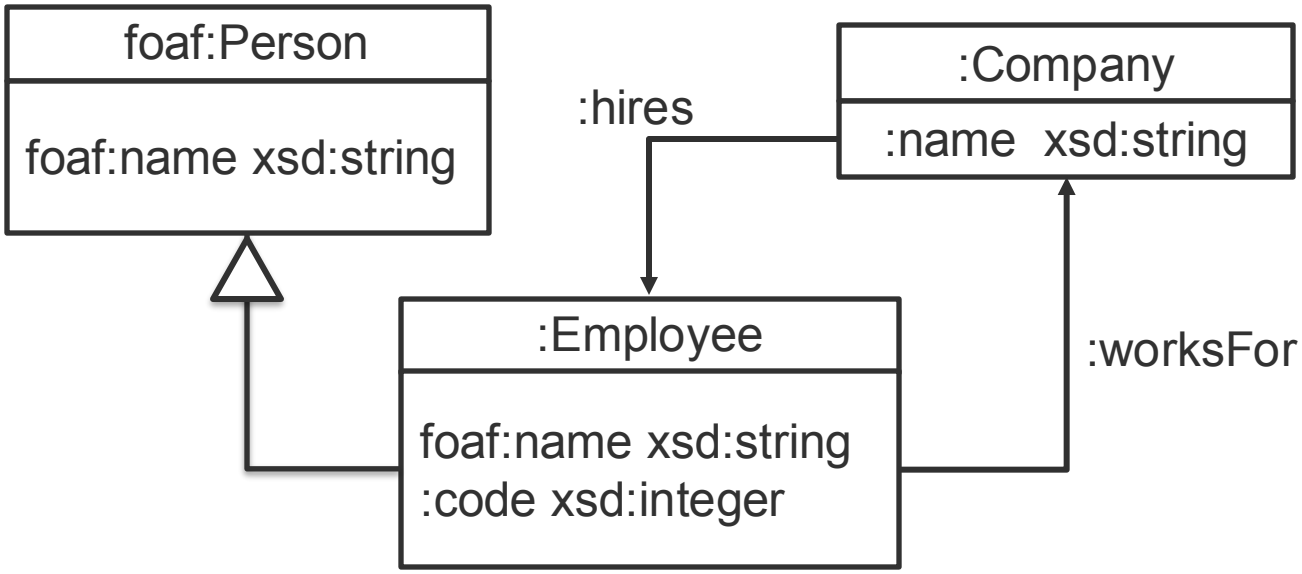
dat:Person2 a foaf:Person;
        foaf:name “Pablo”@es, “Paul”@en;
        foaf:knowns dat:Person21;
       voc:worksIn dat:Org1 .

dat:Org1 a foaf:Organization;
        dat:name “Cantera de Piedradura” .

voc:worksIn a owl:ObjectProperty;
        rdfs:domain foaf:Person; rdfs:range foaf:Organization .
```

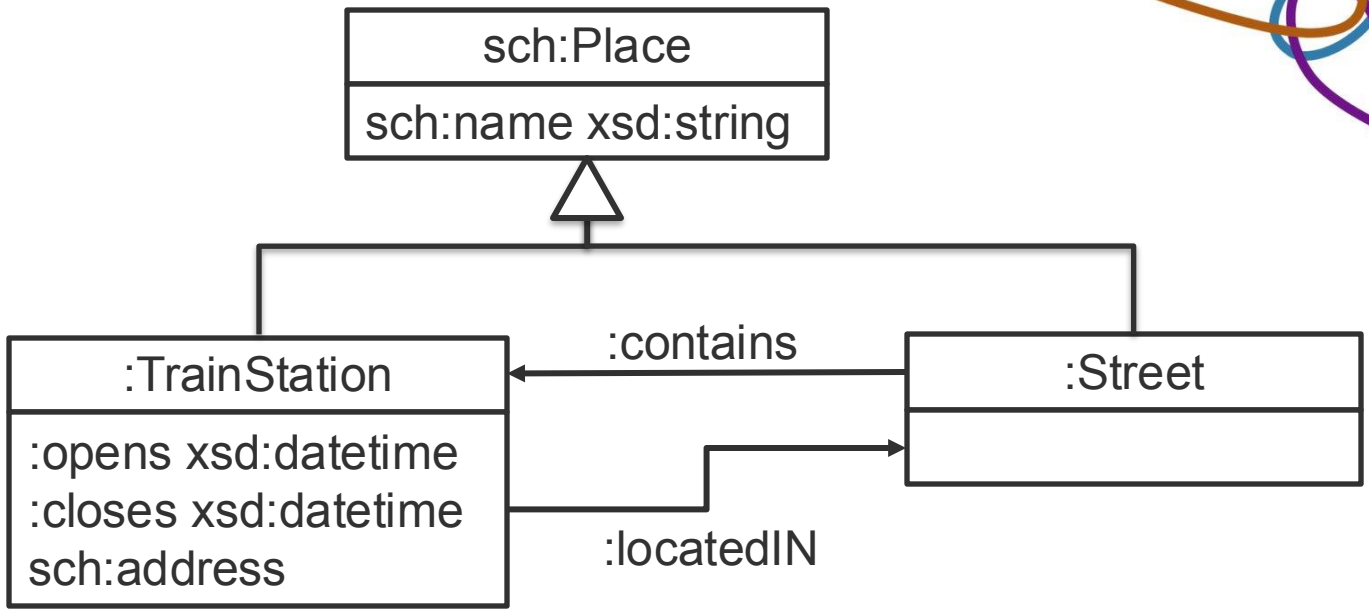
Case 4

- Implement the ontology that conforms to the following UML model. Re-use foaf:Person.



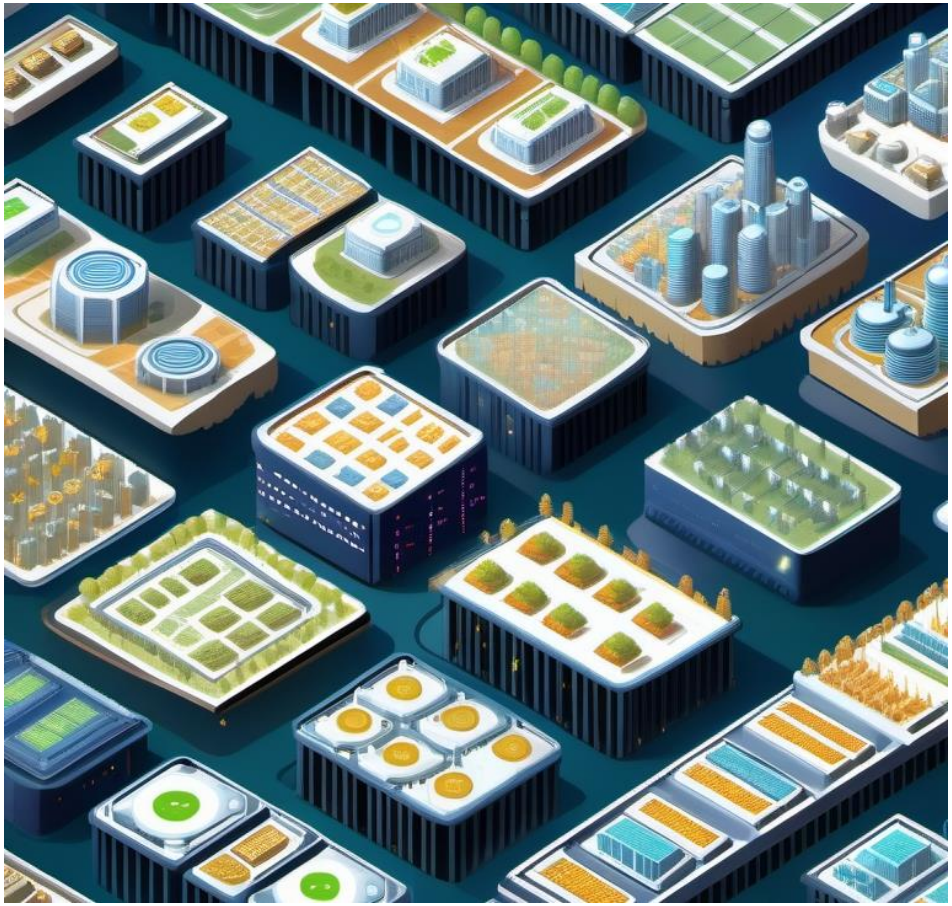
Case 5

- Implement the ontology that conforms to the following UML model.Re-use foaf:Person.Re-use sch:Place.



Case 6

- Writing data using the case study ontology 4
- Writing data using the case study ontology 5



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